

SOFTWARE REQUIREMENTS SPECIFICATION

Inquisitors Society

3D Intelligent Learning & Career Development Platform



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Dated 25th July 2026

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of the **Inquisitors Society 3D Intelligent Learning & Career Development Platform**. The purpose of this document is to define all functional and non-functional requirements, system architecture, user interfaces, and development roadmap for the complete digital transformation of the Inquisitors Society.

The platform aims to create an intelligent educational ecosystem that integrates Learning Management System (LMS), Internship Management, Event Management, Career Development, and Community features with an immersive 3D user experience. This document is intended for:

- **Development Team:** To understand the complete system requirements and architecture
- **Project Managers:** To plan sprints, allocate resources, and track progress
- **Stakeholders:** To validate that all requirements have been captured
- **Quality Assurance:** To create test cases and validate system functionality
- **System Architects:** To ensure the architecture meets all requirements

1.2 Scope

The **Inquisitors Society** is a student-run society at the University of Engineering and Technology, Lahore, dedicated to fostering technical excellence, professional development, and community engagement. Currently, the society organizes:

- Technical Events
- Workshops
- Conferences
- Seminars
- Competitions
- Industrial Tours
- Research Activities
- Career Development Programs
- Leadership Activities
- CSS Sessions
- IELTS Sessions
- Mock Interviews
- Weekly Discussions

- Book Fairs
- Community Activities

The **Inquisitors Society Platform** will digitally transform the society's operations into a complete intelligent education ecosystem. The platform will serve:

- **Students:** Access courses, enroll in events, apply for internships, build portfolios, participate in community discussions
- **Teachers/Faculty:** Create and manage courses, conduct virtual classes, evaluate assignments
- **Mentors:** Guide students, evaluate internship applications, provide feedback
- **Companies:** Post internships, review applications, manage recruitment
- **Administrators:** Manage platform operations, content, users, and analytics

The **system** shall:

1. **Provide a comprehensive Learning Management System** with professional courses, certifications, virtual classrooms, and progress tracking
2. **Enable internship management** from posting to completion with company and mentor oversight
3. **Facilitate event management** including registration, ticketing, QR attendance, and certificate generation
4. **Deliver an immersive 3D user experience** with interactive campus navigation and engaging visual elements
5. **Integrate AI capabilities** for intelligent recommendations, career guidance, and automated assistance
6. **Include community features** for collaboration, discussion, and networking
7. **Provide career development tools** including portfolio building, job board, and skill assessment
8. **Ensure enterprise-level security** with role-based access, encryption, and audit logging

This system is NOT a simple informational website. It is a complete educational ecosystem designed to scale and support thousands of users with enterprise-grade reliability and performance.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
LMS	Learning Management System - Platform for delivering, tracking, and managing educational content
RBAC	Role-Based Access Control - Access restriction based on user roles
JWT	JSON Web Token - Standard for creating access tokens for authentication
API	Application Programming Interface - Set of protocols for software communication
CMS	Content Management System - System for managing digital content
UI/UX	User Interface/User Experience - Design aspects of the platform
DFD	Data Flow Diagram - Visual representation of data flow through a system
ERD	Entity-Relationship Diagram - Visual representation of database structure
GSAP	GreenSock Animation Platform - JavaScript animation library
Three.js	3D JavaScript library for WebGL rendering

Next.js	React framework for server-side rendering and static generation
NestJS	Progressive Node.js framework for building efficient server-side applications
Prisma	Modern ORM (Object-Relational Mapping) for Node.js
WebRTC	Web Real-Time Communication - Framework for peer-to-peer communication
Socket.IO	Library for real-time web applications
AI	Artificial Intelligence - Simulated intelligence in machines
SPA	Single Page Application - Web application that loads a single HTML page
PWA	Progressive Web Application - Web application that behaves like a native app
CI/CD	Continuous Integration/Continuous Deployment - Automation in software development
OAuth	Open Authorization - Standard for token-based authentication

1.4 References

Document/Source	Description
IEEE Std 830-1998	IEEE Recommended Practice for Software Requirements Specifications
ISO/IEC 25010	Systems and Software Engineering - System and Software Quality Models
Inquisitors Society Constitution	Constitution and bylaws of Inquisitors Society
UET Lahore Academic Calendar	University academic calendar and policies
Government of Punjab Education Policy	Provincial education guidelines
OWASP Top 10	Web Application Security Standards
WCAG 2.1	Web Content Accessibility Guidelines

1.5 Overview

The remainder of this document is organized as follows:

- **Section 2:** Overall Description - Provides a general description of the product
- **Section 3:** Stakeholders and User Personas - Identifies stakeholders and defines user types
- **Section 4:** System Features and Functional Requirements - Detailed functional requirements
- **Section 5:** Non-Functional Requirements - Quality attributes and system-wide constraints
- **Section 6:** Business Rules - Rules governing business operations
- **Section 7:** External Interfaces - System interfaces with external systems
- **Section 8:** System Architecture - Technical architecture and technology stack
- **Section 9:** Database Design - Comprehensive database schema
- **Section 10:** Use Cases - Detailed use case specifications
- **Section 11:** Acceptance Criteria - Criteria for accepting the system
- **Section 12:** Risk Analysis - Potential risks and mitigation strategies
- **Section 13:** Future Scope - Planned future enhancements
- **Section 14:** Development Roadmap - Complete project development plan

2. Overall Description

2.1 Product Perspective

The **Inquisitors Society Platform** is a new, self-contained system that will replace the current manual and fragmented processes of the society. It integrates multiple subsystems into a unified platform:

- **Landing Website:** The public-facing website providing access to all platform features
- **Learning Management System:** Educational content delivery and tracking
- **Virtual Classroom:** Live and recorded learning environment
- **Internship Management:** End-to-end internship workflow
- **Event Management:** Complete event lifecycle management
- **Community Forum:** Discussion and collaboration platform
- **Career Portal:** Career development and job matching
- **Admin Dashboard:** Centralized management and monitoring

The platform will be accessible via:

- Web browser (desktop and tablet)
- Mobile-responsive web application
- Progressive Web App (PWA) capabilities

2.2 Product Functions

The platform provides the following high-level functions:

2.2.1 User Management

- Registration and authentication for all user types
- Role-based access control (Student, Teacher, Mentor, Company, Admin)
- Profile management and user settings
- Password recovery and security management

2.2.2 Learning Management System

- Course creation, management, and delivery
- Module and lesson organization
- Assignment submission and grading

- Quizzes and assessments
- Progress tracking and reporting
- Certificate generation with QR verification

2.2.3 Virtual Classroom

- Google Classroom-style class creation
- Live video sessions with Google Meet/Zoom integration
- Resource sharing (videos, documents, PPT)
- Attendance tracking
- Discussion forums and announcements

2.2.4 Internship Management

- Company registration and profile management
- Internship posting and management
- Student application and tracking
- Mentor evaluation workflow
- Internship completion and certificate generation

2.2.5 Event Management

- Event creation and promotion
- Student registration and payment
- QR code ticketing system
- QR attendance verification
- Certificate generation for participants
- Volunteer and organizer management

2.2.6 Career Development

- Student portfolio creation
- Project showcase
- Resume builder with AI assistance
- Job board with company listings
- Skill gap analysis

2.2.7 Community Features

- Discussion forums by topic/category
- Q&A system
- Announcements
- Mentorship matching

2.2.8 AI Module

- AI Chatbot for instant support
- AI Resume Analyzer
- AI Career Advisor
- AI Course and Internship Recommendations
- AI Skill Gap Analysis
- AI Interview Practice
- AI Assignment Feedback

2.2.9 3D User Experience

- Immersive 3D landing scene
- Interactive campus with explorable buildings
- Animated course cards and career roadmap
- Virtual auditorium
- Glassmorphism and modern UI effects

2.3 User Characteristics

2.3.1 Students

- University students seeking courses, events, and career opportunities
- Primary audience with varying technical proficiency
- Mobile-first users who expect intuitive interfaces
- Need for quick access to learning materials and deadlines

2.3.2 Teachers/Faculty

- Subject matter experts delivering educational content
- Technical proficiency varies from basic to advanced
- Need simple tools for course management and student evaluation

2.3.3 Mentors

- Industry professionals guiding and evaluating students
- Time-constrained, need efficient workflows
- Focused on providing quality feedback

2.3.4 Companies

- Organizations posting internships and recruiting talent
- Require structured application management
- Need compliance and reporting features

2.3.5 Administrators

- Platform managers with technical proficiency
- Need comprehensive monitoring and management tools
- Responsible for content approval and user management

2.4 Operating Environment

2.4.1 Client Environment

- Web browsers: Chrome (latest), Firefox (latest), Safari (latest), Edge (latest)
- Mobile browsers: Android Chrome, iOS Safari
- Operating systems: Windows 10+, macOS 10.15+, Linux (Ubuntu 20.04+)
- Internet connection: Minimum 2 Mbps recommended for 3D features

2.4.2 Server Environment

- Cloud deployment (AWS/GCP/Azure)
- Containerized microservices (Docker/Kubernetes)
- Linux server environment (Ubuntu 22.04 LTS)
- PostgreSQL database with read replicas
- Redis for caching and session management
- Object storage (AWS S3 or equivalent) for media files

2.4.3 Development Environment

- Git version control with GitHub/GitLab
- CI/CD pipeline (GitHub Actions or Jenkins)
- Containerized development environment

- Code quality tools (ESLint, Prettier, SonarQube)

2.5 Design and Implementation Constraints

2.5.1 Technical Constraints

- Must be built as a Single Page Application (SPA) using Next.js
- Backend must be built with NestJS and Node.js
- Database must use PostgreSQL with Prisma ORM
- Must be responsive with mobile-first design
- Must implement 3D features using Three.js
- Must use modern animation libraries (GSAP, Framer Motion)
- Must be containerized for deployment
- Must support HTTPS only
- Must be compliant with GDPR and local data protection regulations

2.5.2 Economic Constraints

- Budget-limited project (government/university funded)
- Must use open-source technologies where possible
- Must minimize hardware requirements
- Must optimize for cost-effective cloud deployment

2.5.3 Regulatory Constraints

- Must comply with HEC (Higher Education Commission) standards
- Must comply with Government of Punjab education policies
- Must comply with data protection regulations
- Must ensure accessibility standards (WCAG 2.1 AA)

2.5.4 Platform Constraints

- Must support desktop, tablet, and mobile browsers
- Must be optimized for low-end devices
- Must support offline capabilities (PWA)
- Must support major browsers

2.6 Assumptions and Dependencies

2.6.1 Assumptions

- Internet connectivity will be available for platform access
- Users will have modern browsers with JavaScript enabled
- Users will have basic computer literacy
- Society will provide content and administrative support
- Payment gateway integration will be available
- Google Meet/Zoom integration will be possible
- The university will provide necessary infrastructure support

2.6.2 Dependencies

- Third-party APIs: Payment gateway, Google Meet, Zoom
- Cloud service providers: AWS/GCP/Azure
- External systems: University authentication (optional SSO)
- External services: Email delivery (SendGrid/SES), SMS gateway
- External libraries: Three.js, GSAP, Framer Motion

3. Stakeholders and User Personas

3.1 Primary Stakeholders

Stakeholder	Requirement Gathering Method	Justification
1) Students	I. User Observation II. Surveys and Questionnaires III. Focus Groups Best Method: Surveys	I. User observation helps understand how students currently interact with society activities. II. Surveys reach a large number of students to gather comprehensive requirements. III. Focus groups provide detailed qualitative feedback on desired features. Justification: Surveys are the most efficient method to gather requirements from a large, diverse student body.
2) Teachers	I. Interviews II. Document Analysis III. Workshops Best Method: Interviews	I. Interviews provide deep understanding of teaching needs and workflows. II. Document analysis of existing course materials reveals content requirements. III. Workshops enable collaborative requirement validation.

		Justification: Interviews allow teachers to explain their teaching methodology and how the platform can support it.
3) Mentors	I. Interviews II. Use Case Scenarios III. Surveys Best Method: Interviews	I. Interviews help understand mentor workflows and evaluation needs. II. Use case scenarios show how mentors will interact with the internship system. III. Surveys gather preferences from multiple mentors. Justification: Interviews provide detailed insights into mentor expectations and constraints.
4) Companies	I. Interviews II. Document Analysis III. Prototyping Best Method: Interviews	I. Interviews clarify hiring needs and application management requirements. II. Document analysis of existing recruitment processes reveals gaps. III. Prototyping validates system design with company representatives. Justification: Direct interviews uncover specific requirements for internship and recruitment processes.

3.2 Secondary Stakeholders

Stakeholder	Requirement Gathering Method	Justification
5) System Administrators	I. Interviews II. Workshops III. Document Analysis Best Method: Interviews	I. Interviews uncover system management and monitoring needs. II. Workshops help define security and compliance requirements. III. Document analysis of IT policies ensures alignment. Justification: Administrators have deep system-level knowledge and can articulate infrastructure requirements.
6) Society Executive Council	I. Workshops II. Interviews III. Document Analysis Best Method: Workshops	I. Workshops bring together diverse perspectives on society goals. II. Interviews clarify leadership expectations. III. Document analysis of society constitution reveals governance requirements. Justification: Workshops facilitate collaborative decision-making on strategic requirements.

3.3 Tertiary Stakeholders

Stakeholder	Requirement Gathering Method	Justification
7) UET Lahore Administration	I. Document Analysis II. Interviews Best Method: Document Analysis	I. Document analysis of university policies reveals compliance requirements. II. Interviews clarify high-level expectations. Justification: University requirements are well-documented in official policies.
8) Higher Education Commission (HEC)	I. Document Analysis II. Compliance Review Best Method: Document Analysis	I. HEC guidelines and standards are publicly available. Justification: Requirements are based on established national education standards.
9) Government of Punjab	I. Document Analysis II. Policy Review	I. Government education policies dictate mandatory features.

	Best Method: Document Analysis	Justification: Policy documents provide comprehensive requirements.
10) Employers/Recruiters	I. Interviews II. Surveys Best Method: Surveys	I. Surveys gather broad industry feedback on talent needs. II. Interviews provide detailed insights. Justification: Surveys reach a wide range of employers efficiently.

3.4 User Personas

Persona 1: Ahmed - Student

- **Age:** 20, **Gender:** Male
- **Major:** Computer Science, **Year:** 3rd Year
- **Goals:** Gain technical skills, participate in hackathons, find internships
- **Technical Level:** Intermediate
- **Device:** Primarily smartphone, occasionally laptop
- **Key Needs:** Easy course discovery, mobile-friendly interface, career opportunities

Persona 2: Ayesha - Student/Active Member

- **Age:** 21, **Gender:** Female
- **Major:** Software Engineering, **Year:** Final Year
- **Goals:** Complete courses, participate in research, secure job
- **Technical Level:** Advanced
- **Device:** Laptop and smartphone
- **Key Needs:** Quality content, portfolio building, networking opportunities

Persona 3: Dr. Hassan - Faculty

- **Age:** 45, **Gender:** Male
- **Position:** Associate Professor, **Department:** Computer Science
- **Goals:** Deliver effective online courses, assess students efficiently
- **Technical Level:** Moderate
- **Device:** Laptop, occasionally tablet
- **Key Needs:** Simple course management, reliable video delivery, grading tools

Persona 4: Fatima - Mentor

- **Age:** 30, **Gender:** Female
- **Position:** Software Engineer (Industry)

- **Goals:** Guide students, evaluate internship applications
- **Technical Level:** High
- **Device:** Laptop (work), smartphone
- **Key Needs:** Efficient evaluation workflow, clear student profiles

Persona 5: TechCorp - Company HR

- **Representative:** Mr. Bilal
- **Company:** TechCorp Solutions
- **Goals:** Post internships, find qualified students, streamline recruitment
- **Technical Level:** Moderate
- **Device:** Desktop (work)
- **Key Needs:** Easy internship posting, application filtering, communication tools

Persona 6: Mr. Khalid - System Admin

- **Age:** 35, **Gender:** Male
- **Position:** IT Manager, UET Lahore
- **Goals:** Ensure platform stability, security, and compliance
- **Technical Level:** Advanced
- **Device:** Multiple (desktop, laptop, server)
- **Key Needs:** Monitoring dashboards, backup systems, audit logs

4. System Features and Functional Requirements

4.1 User Management and Authentication Module

4.1.1 Student/User Registration

ID	Requirement
FR-01-001	The system shall allow new users to register using email, phone number, or Google OAuth.
FR-01-002	The system shall validate email addresses and phone numbers during registration.
FR-01-003	The system shall require users to verify their email address before accessing the platform.

FR-01-004	The system shall collect basic user information: Full Name, CNIC/Passport, Date of Birth, Gender, Department, Year of Study, University/Institute.
FR-01-005	The system shall allow users to select their role during registration (Student, Teacher, Mentor, Company).
FR-01-006	The system shall provide multi-step registration for companies (Company Name, Registration Number, Industry, Contact Person).
FR-01-007	The system shall require users to accept Terms of Service and Privacy Policy during registration.
FR-01-008	The system shall allow users to complete profile setup after registration (Profile Photo, Bio, Skills, Interests).
FR-01-009	The system shall send welcome email upon successful registration.
FR-01-010	The system shall support university email domain validation for student and teacher accounts.

4.1.2 Authentication

ID	Requirement
FR-01-011	The system shall support email/password authentication with JWT token generation.
FR-01-012	The system shall support Google OAuth authentication.
FR-01-013	The system shall support GitHub OAuth authentication.
FR-01-014	The system shall provide "Remember Me" functionality with secure token storage.
FR-01-015	The system shall support "Forgot Password" with OTP-based recovery via email.
FR-01-016	The system shall require users to verify their email address for password recovery.
FR-01-017	The system shall implement session management with automatic logout after inactivity.
FR-01-018	The system shall support multi-factor authentication (MFA) for admin accounts.
FR-01-019	The system shall log all successful and failed login attempts.
FR-01-020	The system shall lock accounts after 5 failed login attempts and require admin unlock.

4.1.3 Profile Management

ID	Requirement
FR-01-021	The system shall allow users to view and edit their profile information.
FR-01-022	The system shall allow users to upload and update profile pictures (1MB limit, JPEG/PNG).
FR-01-023	The system shall allow users to add/update their skills and professional interests.
FR-01-024	The system shall allow students to manage academic information (Courses, GPA, Year).
FR-01-025	The system shall allow students to upload CV, Portfolio, GitHub, LinkedIn links.
FR-01-026	The system shall allow companies to manage company profile (Description, Logo, Website, Industry).
FR-01-027	The system shall allow users to change their password with current password verification.
FR-01-028	The system shall allow users to update notification preferences.
FR-01-029	The system shall allow users to delete their account with confirmation and data retention policy.
FR-01-030	The system shall provide public profile view for students (visible to companies and mentors).

4.1.4 Role-Based Access Control

ID	Requirement
----	-------------

FR-01-031	The system shall implement RBAC with the following roles: Student, Teacher, Mentor, Company, Admin.
FR-01-032	The system shall restrict access to features based on user role.
FR-01-033	The system shall allow admins to assign roles to users.
FR-01-034	The system shall allow admins to modify user permissions.
FR-01-035	The system shall log all access attempts to restricted resources.
FR-01-036	The system shall display appropriate UI elements based on user role.
FR-01-037	The system shall support hierarchical role permissions (Admin > Teacher > Mentor > Student).
FR-01-038	The system shall allow temporary role assignments for event organizers or volunteers.

4.2 Learning Management System (LMS)

4.2.1 Course Management

ID	Requirement
FR-02-001	The system shall allow teachers to create courses with basic information (Title, Description, Category, Learning Objectives, Prerequisites).
FR-02-002	The system shall allow teachers to set course visibility (Public, Private (Unlisted), Restricted (Members Only)).
FR-02-003	The system shall allow admins to approve courses before publication.
FR-02-004	The system shall support course categories and subcategories (Technology, Business, Design, Language, etc.).
FR-02-005	The system shall display a course catalog with search and filtering capabilities.
FR-02-006	The system shall allow users to enroll in courses with a single click.
FR-02-007	The system shall support course pricing (Free or Paid).
FR-02-008	The system shall allow teachers to provide course updates and announcements.
FR-02-009	The system shall display course ratings and reviews.
FR-02-010	The system shall support course duplicates or templates for recurring courses.
FR-02-011	The system shall allow course instructors to be added (multiple instructors per course).
FR-02-012	The system shall support course drafts for instructors working on future courses.

4.2.2 Lesson Management

ID	Requirement
FR-02-013	The system shall allow teachers to create modules/sections within a course.

FR-02-014	The system shall allow teachers to create lessons within modules.
FR-02-015	The system shall support the following lesson types: Video (Recorded), Video (Live), Text/PDF, Assignment, Quiz, Coding Lab, Presentation.
FR-02-016	The system shall allow teachers to upload video files with automatic transcoding.
FR-02-017	The system shall support video streaming with adaptive bitrate.
FR-02-018	The system shall allow teachers to embed YouTube/Vimeo videos.
FR-02-019	The system shall support PDF, Word, PPT, Excel uploads with preview capabilities.
FR-02-020	The system shall allow teachers to set lesson durations (estimated time to complete).
FR-02-021	The system shall support lesson prerequisites (must complete before accessing next).
FR-02-022	The system shall track student progress per lesson.
FR-02-023	The system shall allow teachers to reorder lessons within modules.
FR-02-024	The system shall allow teachers to add downloadable resources (Notes, Code files, etc.).
FR-02-025	The system shall support lesson comments and questions.

4.2.3 Assignment Management

ID	Requirement
FR-02-026	The system shall allow teachers to create assignments with clear instructions.
FR-02-027	The system shall support assignment deadlines and time limits.
FR-02-028	The system shall allow file uploads for assignment submissions.
FR-02-029	The system shall support multiple submission attempts (with teacher configuration).
FR-02-030	The system shall allow teachers to provide feedback and grades on assignments.
FR-02-031	The system shall support plagiarism detection (integration with external service).
FR-02-032	The system shall allow teachers to create coding assignments with live code editors.
FR-02-033	The system shall support assignment late submission policies.
FR-02-034	The system shall notify students of assignment grades and feedback.
FR-02-035	The system shall allow teachers to create grading rubrics.
FR-02-036	The system shall support peer review assignments.
FR-02-037	The system shall store all assignment submissions for future reference.

4.2.4 Quiz Management

ID	Requirement
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FR-02-038	The system shall allow teachers to create quizzes with multiple question types (Multiple Choice, True/False, Fill in the Blank, Matching, Essay).
FR-02-039	The system shall support question banks that can be shared across courses.
FR-02-040	The system shall allow randomizing question order.
FR-02-041	The system shall support time limits on quizzes.
FR-02-042	The system shall support automated grading for objective questions.
FR-02-043	The system shall allow manual grading for subjective/essay questions.
FR-02-044	The system shall provide instant feedback on correct/incorrect answers.
FR-02-045	The system shall support passing scores and certificate eligibility.
FR-02-046	The system shall display quiz results to students immediately after completion.
FR-02-047	The system shall allow teachers to review and override quiz scores.
FR-02-048	The system shall provide quiz analytics (pass rates, average scores, question difficulty).
FR-02-049	The system shall allow teachers to create practice quizzes without grading.
FR-02-050	The system shall support AI-generated quiz questions based on course content (see AI module).

4.2.5 Progress Tracking

ID	Requirement
FR-02-051	The system shall track each student's progress through course content.
FR-02-052	The system shall display a progress bar for each course.
FR-02-053	The system shall mark lessons as complete automatically when viewed.
FR-02-054	The system shall allow students to manually mark lessons complete.
FR-02-055	The system shall display estimated time remaining to complete a course.
FR-02-056	The system shall show completion certificates when course requirements are met.
FR-02-057	The system shall display course completion status on student dashboard.
FR-02-058	The system shall allow teachers to view student progress reports.
FR-02-059	The system shall provide analytics on course completion rates.
FR-02-060	The system shall support learning paths with sequential courses.

4.2.6 Certificate Generation

ID	Requirement
FR-02-061	The system shall generate certificates for course completion.

FR-02-062	The system shall generate certificates for event participation.
FR-02-063	The system shall generate certificates for internship completion.
FR-02-064	The system shall support customizable certificate templates.
FR-02-065	The system shall include a unique QR code on each certificate.
FR-02-066	The system shall allow QR code verification for certificate authenticity.
FR-02-067	The system shall allow students to download certificates in PDF format.
FR-02-068	The system shall allow students to share certificates on social media.
FR-02-069	The system shall store certificates in a public directory for verification.
FR-02-070	The system shall include certificate metadata (Issue Date, Expiry Date, Course/Event Details).
FR-02-071	The system shall allow teachers to approve certificate eligibility.
FR-02-072	The system shall prevent duplicate certificates for the same achievement.

4.3 Virtual Classroom Module

4.3.1 Class Creation

ID	Requirement
FR-03-001	The system shall allow teachers to create virtual classrooms.
FR-03-002	The system shall require class details: Name, Description, Subject, Grade Level, Section.
FR-03-003	The system shall support class roster management (enroll students, remove students).
FR-03-004	The system shall provide unique class code for student enrollment.
FR-03-005	The system shall allow teachers to create multiple classes per semester.
FR-03-006	The system shall support class archives at the end of semester.
FR-03-007	The system shall allow teachers to add co-teachers/assistants.
FR-03-008	The system shall automatically generate class calendar.

4.3.2 Live Sessions

ID	Requirement
FR-03-009	The system shall support live video sessions through Google Meet integration.
FR-03-010	The system shall support live video sessions through Zoom integration.
FR-03-011	The system shall allow teachers to schedule live sessions.
FR-03-012	The system shall send reminders to students before scheduled sessions.
FR-03-013	The system shall allow recording of live sessions.
FR-03-014	The system shall store session recordings in the class content.
FR-03-015	The system shall support screen sharing during live sessions.
FR-03-016	The system shall support live chat and Q&A during sessions.
FR-03-017	The system shall support breakout rooms for group activities.
FR-03-018	The system shall allow teachers to manage participant permissions.

4.3.3 Attendance Management

ID	Requirement
FR-03-019	The system shall automatically record attendance during live sessions.
FR-03-020	The system shall allow teachers to manually mark attendance.
FR-03-021	The system shall display attendance reports to teachers.

FR-03-022	The system shall allow students to view their attendance records.
FR-03-023	The system shall support QR code-based attendance check-in.
FR-03-024	The system shall generate attendance warnings for students with low attendance.
FR-03-025	The system shall support attendance tracking for multiple sessions.

4.3.4 Discussion Forums

ID	Requirement
FR-03-026	The system shall provide discussion forums for each classroom.
FR-03-027	The system shall allow students to create new discussion threads.
FR-03-028	The system shall allow teachers to post announcements as sticky threads.
FR-03-029	The system shall support comment nesting on discussion threads.
FR-03-030	The system shall allow users to like and reply to comments.
FR-03-031	The system shall support file attachments in discussions.
FR-03-032	The system shall send notifications for new comments.
FR-03-033	The system shall allow teachers to moderate discussions.

4.3.5 Announcements

ID	Requirement
FR-03-034	The system shall allow teachers to post announcements to classrooms.
FR-03-035	The system shall support multiple announcement formats (Text, Video, Link).
FR-03-036	The system shall send push notifications for new announcements.
FR-03-037	The system shall send email notifications for important announcements.
FR-03-038	The system shall allow announcements to be saved as drafts.
FR-03-039	The system shall show read receipts for announcements.
FR-03-040	The system shall allow teachers to schedule announcements.

4.4 Internship Management Module

4.4.1 Internship Posting

ID	Requirement
FR-04-001	The system shall allow companies to register and create company profiles.
FR-04-002	The system shall allow companies to post internships with details: Title, Description, Requirements, Duration, Stipend, Location, Number of Positions.
FR-04-003	The system shall require admin approval for new company registrations.
FR-04-004	The system shall require admin approval for internship postings.
FR-04-005	The system shall allow internships to be categorized by industry, skills, duration.
FR-04-006	The system shall allow companies to edit internship listings.

FR-04-007	The system shall allow companies to close internship applications early.
FR-04-008	The system shall send expiration notifications for internships.
FR-04-009	The system shall support internship types (Paid, Unpaid, Academic Credit).
FR-04-010	The system shall allow companies to list required skills and preferences.
FR-04-011	The system shall support internship templates for recurring internships.

4.4.2 Student Application

ID	Requirement
FR-04-012	The system shall allow students to view available internships.
FR-04-013	The system shall allow students to search and filter internships.
FR-04-014	The system shall allow students to view internship details and requirements.
FR-04-015	The system shall allow students to apply for internships with a single click.
FR-04-016	The system shall allow students to upload CV during application.
FR-04-017	The system shall allow students to provide cover letters with applications.
FR-04-018	The system shall allow students to link GitHub, LinkedIn, Portfolio.
FR-04-019	The system shall track application status (Applied, Under Review, Shortlisted, Selected, Rejected).
FR-04-020	The system shall notify students of application status changes.
FR-04-021	The system shall support multiple internship applications.
FR-04-022	The system shall allow students to withdraw applications before review.

4.4.3 Mentor Evaluation

ID	Requirement
FR-04-023	The system shall allow mentors to review student applications.
FR-04-024	The system shall display student profiles and application materials.
FR-04-025	The system shall allow mentors to evaluate students with scoring metrics.
FR-04-026	The system shall allow mentors to provide feedback on student applications.
FR-04-027	The system shall allow mentors to shortlist candidates.
FR-04-028	The system shall support mentor comments and notes on applications.
FR-04-029	The system shall allow mentors to request additional information from students.
FR-04-030	The system shall provide mentors with evaluation dashboards.

4.4.4 Approval Workflow

ID	Requirement
FR-04-031	The system shall support multi-step approval workflow for internships.
FR-04-032	The system shall require society approval before internships are posted.
FR-04-033	The system shall require admin approval for student selection.
FR-04-034	The system shall support email notifications for each workflow step.
FR-04-035	The system shall maintain approval history and audit logs.

FR-04-036	The system shall allow designated approvers to act on pending items.
FR-04-037	The system shall support re-assignment of approvers.
FR-04-038	The system shall allow comments at each approval step.

4.4.5 Internship Reporting

ID	Requirement
FR-04-039	The system shall allow students to submit internship reports.
FR-04-040	The system shall allow students to upload weekly logs.
FR-04-041	The system shall allow mentors to review student performance during internship.
FR-04-042	The system shall allow mentors to provide final feedback and evaluation.
FR-04-043	The system shall track student attendance during internship.
FR-04-044	The system shall allow companies to provide feedback on students.
FR-04-045	The system shall generate internship completion certificates.
FR-04-046	The system shall provide performance analytics for internships.
FR-04-047	The system shall support internship evaluation forms.
FR-04-048	The system shall store internship reports for future reference.
FR-04-049	The system shall allow students to include internships in their portfolio.

4.5 Event Management Module

4.5.1 Event Creation

ID	Requirement
FR-05-001	The system shall allow society admins and event organizers to create events.
FR-05-002	The system shall support event types: Seminar, Workshop, Hackathon, Conference, Competition, Industrial Tour, Book Fair, Research Conference, Leadership Session, CSS Session, IELTS Session.
FR-05-003	The system shall require event details: Title, Description, Date, Time, Venue, Capacity, Registration Deadline, Event Type, Category.
FR-05-004	The system shall allow event organizers to set event visibility (Public, Society Members Only, Invitation Only).
FR-05-005	The system shall support event images and promotional materials.
FR-05-006	The system shall allow event organizers to set registration fees (Free or Paid).
FR-05-007	The system shall allow event organizers to add speakers or guests.
FR-05-008	The system shall generate event schedules/agendas.
FR-05-009	The system shall allow event organizers to send event invitations.
FR-05-010	The system shall support recurring events.

4.5.2 Registration and Payment

ID	Requirement
FR-05-011	The system shall allow students to register for events.
FR-05-012	The system shall integrate with payment gateway for paid events.
FR-05-013	The system shall support multiple payment methods (Credit/Debit Card, JazzCash, EasyPaiza).
FR-05-014	The system shall generate payment receipts.
FR-05-015	The system shall support refund policies and processing.
FR-05-016	The system shall track registration status (Pending, Confirmed, Waitlisted, Cancelled).
FR-05-017	The system shall limit registrations to event capacity.
FR-05-018	The system shall support group registrations.
FR-05-019	The system shall display registration count and remaining slots.
FR-05-020	The system shall allow event organizers to export attendee lists.

4.5.3 QR Code Ticketing

ID	Requirement
FR-05-021	The system shall generate unique QR codes for each registered participant.
FR-05-022	The system shall allow participants to view/download tickets with QR code.
FR-05-023	The system shall support QR code scanning for entry verification.
FR-05-024	The system shall mark attendance when QR code is scanned.
FR-05-025	The system shall prevent duplicate scanning of the same QR code.
FR-05-026	The system shall allow event organizers to scan QR codes via mobile app.
FR-05-027	The system shall provide real-time attendance tracking.
FR-05-028	The system shall generate attendance reports.
FR-05-029	The system shall support offline QR scanning with sync.

4.5.4 Volunteer Management

ID	Requirement
FR-05-030	The system shall allow students to register as volunteers for events.
FR-05-031	The system shall allow event organizers to assign volunteer roles.
FR-05-032	The system shall support volunteer shift scheduling.
FR-05-033	The system shall track volunteer hours.
FR-05-034	The system shall generate volunteer certificates.
FR-05-035	The system shall allow event organizers to add students as event organizers.
FR-05-036	The system shall support organizer permissions and responsibilities.
FR-05-037	The system shall allow event organizers to communicate with volunteers.

4.5.5 Gallery Management

ID	Requirement
FR-05-038	The system shall allow event organizers to upload event photos.
FR-05-039	The system shall allow event organizers to upload event videos.
FR-05-040	The system shall display event galleries on the platform.
FR-05-041	The system shall support image optimization for fast loading.

FR-05-042	The system shall allow users to download images (if permitted).
FR-05-043	The system shall support album creation for each event.
FR-05-044	The system shall display highlights of recent events on landing page.
FR-05-045	The system shall support social media sharing of event media.

4.6 Career Development Module

4.6.1 Job Board

ID	Requirement
FR-06-001	The system shall allow companies to post job opportunities.
FR-06-002	The system shall allow students to search and apply for jobs.
FR-06-003	The system shall provide job filters by industry, location, experience level.
FR-06-004	The system shall display job recommendations based on student profile.
FR-06-005	The system shall allow students to save jobs for later application.
FR-06-006	The system shall provide application tracking for students.
FR-06-007	The system shall support company profiles with reviews and ratings.
FR-06-008	The system shall allow companies to manage job applications.
FR-06-009	The system shall send job alert notifications.
FR-06-010	The system shall support job expiration and renewal.

4.6.2 Portfolio Management

ID	Requirement
FR-06-011	The system shall allow students to create online portfolios.
FR-06-012	The system shall support portfolio sections: About Me, Education, Experience, Projects, Skills, Certificates, Publications.
FR-06-013	The system shall allow students to upload project files, images, and videos.
FR-06-014	The system shall allow students to customize portfolio design.
FR-06-015	The system shall generate public portfolio URLs.
FR-06-016	The system shall support portfolio sharing on social media.
FR-06-017	The system shall allow students to showcase achievements.
FR-06-018	The system shall provide portfolio analytics (views, engagement).

4.6.3 Project Showcase

ID	Requirement
FR-06-019	The system shall allow students to showcase their projects.

FR-06-020	The system shall support project details: Title, Description, Technologies Used, Team Members, GitHub Link, Live Demo.
FR-06-021	The system shall allow project images and video demos.
FR-06-022	The system shall display projects in a gallery format.
FR-06-023	The system shall allow likes and comments on projects.
FR-06-024	The system shall support project categories.
FR-06-025	The system shall feature top projects on landing page.

4.6.4 Resume Builder

ID	Requirement
FR-06-026	The system shall provide a resume builder tool.
FR-06-027	The system shall support multiple resume templates.
FR-06-028	The system shall allow students to import profile data into resume.
FR-06-029	The system shall support download in PDF format.
FR-06-030	The system shall allow students to create multiple resumes.
FR-06-031	The system shall provide AI-assisted resume content suggestions.
FR-06-032	The system shall support resume editing and preview.

4.7 Community Forum Module

4.7.1 Discussion Threads

ID	Requirement
FR-07-001	The system shall provide community forums for discussions.
FR-07-002	The system shall support forum categories (General, Technical, Career, Events, Research, Leadership).
FR-07-003	The system shall allow users to create discussion threads.
FR-07-004	The system shall allow users to reply to threads.
FR-07-005	The system shall support nested comments.
FR-07-006	The system shall allow likes and upvotes on comments.
FR-07-007	The system shall support best answer marking.
FR-07-008	The system shall provide search functionality for forums.
FR-07-009	The system shall allow users to follow threads.
FR-07-010	The system shall support thread tagging and categorization.
FR-07-011	The system shall display active threads on the dashboard.

4.7.2 Q&A System

ID	Requirement
FR-07-012	The system shall support a Q&A section for technical questions.
FR-07-013	The system shall allow users to ask questions.

FR-07-014	The system shall allow users to answer questions.
FR-07-015	The system shall support accepted answer marking.
FR-07-016	The system shall provide point/reputation system for helpful answers.
FR-07-017	The system shall support question categories and tags.
FR-07-018	The system shall allow duplicate question detection.
FR-07-019	The system shall send notifications for answered questions.
FR-07-020	The system shall support question voting.

4.7.3 Notification System

ID	Requirement
FR-07-021	The system shall provide real-time notifications for all user activities.
FR-07-022	The system shall support notification types: Course updates, Assignment grades, Event reminders, Internship status, Forum replies, Likes and comments, System announcements.
FR-07-023	The system shall support push notifications (web and mobile).
FR-07-024	The system shall support email notifications.
FR-07-025	The system shall allow users to configure notification preferences.
FR-07-026	The system shall support in-app notifications.
FR-07-027	The system shall mark notifications as read/unread.
FR-07-028	The system shall provide notification history.
FR-07-029	The system shall group related notifications.

4.8 AI Module

4.8.1 AI Chatbot

ID	Requirement
FR-08-001	The system shall provide an AI-powered chatbot for user assistance.
FR-08-002	The chatbot shall answer FAQs about courses, events, internships.
FR-08-003	The chatbot shall guide users through platform features.
FR-08-004	The chatbot shall provide course recommendations.
FR-08-005	The chatbot shall provide event recommendations.
FR-08-006	The chatbot shall support natural language processing for flexible queries.
FR-08-007	The chatbot shall escalate complex queries to human support.
FR-08-008	The chatbot shall learn from user interactions to improve.
FR-08-009	The chatbot shall support multiple languages (English and Urdu).
FR-08-010	The chatbot shall maintain conversation history.

4.8.2 AI Resume Analyzer

ID	Requirement
FR-08-011	The system shall analyze student resumes using AI.
FR-08-012	The system shall provide suggestions for resume improvement.
FR-08-013	The system shall identify missing skills and keywords.
FR-08-014	The system shall score resumes against job requirements.
FR-08-015	The system shall suggest relevant skills to add.
FR-08-016	The system shall provide formatting recommendations.
FR-08-017	The system shall check for common resume mistakes.
FR-08-018	The system shall provide comparative analysis with industry standards.

4.8.3 AI Career Advisor

ID	Requirement
FR-08-019	The system shall provide career guidance based on student profile.
FR-08-020	The system shall recommend career paths based on skills and interests.
FR-08-021	The system shall suggest relevant courses and certifications.
FR-08-022	The system shall identify job market trends in student's field.
FR-08-023	The system shall recommend skill development for career goals.
FR-08-024	The system shall provide career roadmaps with milestones.
FR-08-025	The system shall suggest mentorship and networking opportunities.

4.8.4 AI Course Recommendation

ID	Requirement
FR-08-026	The system shall recommend courses based on student interests and skills.
FR-08-027	The system shall suggest learning paths for career goals.
FR-08-028	The system shall recommend courses based on career trends.
FR-08-029	The system shall consider student's current skill level.
FR-08-030	The system shall adapt recommendations based on progress.

4.8.5 AI Skill Gap Analysis

ID	Requirement
FR-08-031	The system shall analyze student skills and identify gaps.
FR-08-032	The system shall compare student skills with industry requirements.
FR-08-033	The system shall suggest courses to fill skill gaps.
FR-08-034	The system shall provide skill proficiency assessments.
FR-08-035	The system shall track skill development progress.
FR-08-036	The system shall generate skill gap reports.

4.8.6 AI Interview Practice

ID	Requirement
FR-08-037	The system shall provide AI-powered mock interview practice.
FR-08-038	The system shall generate interview questions based on job role.

FR-08-039	The system shall evaluate responses and provide feedback.
FR-08-040	The system shall simulate technical and behavioral interviews.
FR-08-041	The system shall provide interview tips and best practices.
FR-08-042	The system shall track practice progress.
FR-08-043	The system shall support audio/video responses.

4.8.7 AI Quiz Generator

ID	Requirement
FR-08-044	The system shall automatically generate quiz questions from course content.
FR-08-045	The system shall support multiple question types.
FR-08-046	The system shall allow teachers to customize generated questions.
FR-08-047	The system shall provide difficulty levels for questions.
FR-08-048	The system shall generate chapter-based quizzes.
FR-08-049	The system shall generate comprehensive course assessments.
FR-08-050	The system shall allow export of generated quizzes.

4.9 Admin Dashboard

4.9.1 Analytics Dashboard

ID	Requirement
FR-09-001	The system shall provide an analytics dashboard for admins.
FR-09-002	The system shall display user statistics (Total users, Active users, New registrations).
FR-09-003	The system shall display course analytics (Enrollments, Completion rates, Popular courses).
FR-09-004	The system shall display event analytics (Registrations, Attendance, Popular events).
FR-09-005	The system shall display internship analytics (Postings, Applications, Placements).
FR-09-006	The system shall display revenue analytics (Payments, Refunds).
FR-09-007	The system shall display engagement metrics (Forum activity, Session durations).
FR-09-008	The system shall provide customizable date ranges for analytics.
FR-09-009	The system shall export analytics reports.
FR-09-010	The system shall provide real-time system health monitoring.

4.9.2 Content Management

ID	Requirement
FR-09-011	The system shall allow admins to manage all content (Courses, Events, Internships, Jobs).
FR-09-012	The system shall allow admins to approve content submissions.
FR-09-013	The system shall allow admins to feature content on landing page.
FR-09-014	The system shall allow admins to archive or delete outdated content.
FR-09-015	The system shall allow admins to manage content categories.
FR-09-016	The system shall provide content search and filtering.
FR-09-017	The system shall support content versioning.
FR-09-018	The system shall allow bulk content operations.

4.9.3 User Management

ID	Requirement
FR-09-019	The system shall allow admins to view all registered users.
FR-09-020	The system shall allow admins to search and filter users.
FR-09-021	The system shall allow admins to assign/change user roles.
FR-09-022	The system shall allow admins to activate/deactivate accounts.
FR-09-023	The system shall allow admins to delete user accounts (with data retention policy).
FR-09-024	The system shall allow admins to view user activity logs.
FR-09-025	The system shall allow admins to send communications to users.
FR-09-026	The system shall allow admins to manage support tickets.

4.9.4 System Monitoring

ID	Requirement
FR-09-027	The system shall provide system performance monitoring.
FR-09-028	The system shall display server health metrics (CPU, Memory, Disk, Network).
FR-09-029	The system shall monitor API response times.
FR-09-030	The system shall monitor database performance.
FR-09-031	The system shall generate alerts for system anomalies.
FR-09-032	The system shall provide audit logs for all admin actions.
FR-09-033	The system shall support system backup and restore.
FR-09-034	The system shall allow scheduled maintenance notifications.

4.10 3D User Experience Module

4.10.1 3D Landing Scene

ID	Requirement
FR-10-001	The system shall display an immersive 3D hero scene on the landing page.
FR-10-002	The scene shall render a university campus environment with buildings representing society departments.
FR-10-003	The scene shall incorporate a "Knowledge Cube" that rotates and displays course categories.
FR-10-004	The scene shall include floating educational elements (books, graduation caps, gears).
FR-10-005	The scene shall use an aurora/particle background effect.
FR-10-006	The scene shall respond to mouse movement with smooth parallax effects.
FR-10-007	The scene shall load optimized assets for performance.
FR-10-008	The scene shall include interactive hot spots for navigation.
FR-10-009	The scene shall adapt to different screen sizes.

4.10.2 Interactive Campus

ID	Requirement
FR-10-010	The system shall provide interactive campus navigation experience.
FR-10-011	Users shall be able to click on buildings to explore different modules.
FR-10-012	Each building shall represent a module (LMS, Internship, Events, Career, Community, Admin).
FR-10-013	Hovering over buildings shall display tooltips with module names.

FR-10-014	Clicking a building shall transition smoothly to the corresponding section.
FR-10-015	The campus shall include animated elements (trees, clouds, students, vehicles).
FR-10-016	The campus shall include a path/bridge representing career roadmap.

4.10.3 Animated Course Cards

ID	Requirement
FR-10-017	The system shall display courses as animated 3D cards.
FR-10-018	Cards shall flip to reveal course details on hover.
FR-10-019	Cards shall have smooth entrance animations on scroll.
FR-10-020	Cards shall include subtle hover effects.
FR-10-021	Cards shall show progress bars for enrolled courses.
FR-10-022	Cards shall adapt colors based on category.

4.10.4 Achievement Wall

ID	Requirement
FR-10-023	The system shall display an interactive achievement wall.
FR-10-024	Achievements (certificates, medals, trophies) shall be displayed in a 3D gallery.
FR-10-025	Achievements shall have reveal animations on scroll.
FR-10-026	Users shall be able to filter achievements by category.
FR-10-027	Achievements shall include clickable links to certificate verification.

4.10.5 Virtual Auditorium

ID	Requirement
FR-10-028	The system shall provide a virtual auditorium for events and conferences.
FR-10-029	The auditorium shall include a stage with speaker area.
FR-10-030	The auditorium shall display upcoming events.
FR-10-031	The auditorium shall include seating areas that represent registration status.
FR-10-032	The auditorium shall include interactive elements for event navigation.

5. Non-Functional Requirements

5.1 Performance Requirements

ID	Requirement
NFR-001	The system must respond to user interactions within 2 seconds (95th percentile).
NFR-002	The system must load the landing page within 3 seconds on a standard broadband connection.
NFR-003	The system must support at least 10,000 concurrent users without performance degradation.
NFR-004	The system must handle at least 1,000 requests per second on the API layer.
NFR-005	The system must load 3D scenes within 5 seconds with caching enabled.
NFR-006	The system must cache static assets with appropriate headers.
NFR-007	The system must use CDN for static asset delivery.

NFR-008	The system must optimize images and media files for web delivery.
NFR-009	The system must maintain database query response times under 100ms for common operations.
NFR-010	The system must support lazy loading for content below the fold.

5.2 Reliability Requirements

ID	Requirement
NFR-011	The system must maintain 99.9% uptime during operational hours.
NFR-012	The system must ensure 99.99% data integrity across all operations.
NFR-013	The system must handle network interruptions gracefully.
NFR-014	The system must implement automatic retries for failed operations.
NFR-015	The system must not lose data during server failures.
NFR-016	The system must implement automatic database backups.
NFR-017	The system must have zero data loss in transactions.
NFR-018	The system must provide operational logs for troubleshooting.

5.3 Scalability Requirements

ID	Requirement
NFR-019	The system must be horizontally scalable to support growth.
NFR-020	The system must support database read replicas for scaling.
NFR-021	The system must support caching layer (Redis) for performance.
NFR-022	The system must handle increased user load during peak events.
NFR-023	The system must support auto-scaling based on load.
NFR-024	The system must be able to handle 100,000 users within 3 years.
NFR-025	The system must scale file storage to support growing media content.

5.4 Security Requirements

ID	Requirement
NFR-026	The system must encrypt all data at rest using AES-256.
NFR-027	The system must encrypt all data in transit using TLS 1.3.
NFR-028	The system must use JWT for stateless authentication.
NFR-029	The system must implement rate limiting for API endpoints.
NFR-030	The system must validate all user inputs to prevent injection attacks.
NFR-031	The system must hash all passwords using bcrypt.
NFR-032	The system must implement role-based access control (RBAC).
NFR-033	The system must log all security-relevant events.
NFR-034	The system must implement CORS policies strictly.
NFR-035	The system must use HTTPS exclusively.
NFR-036	The system must implement multi-factor authentication for admin accounts.
NFR-037	The system must automatically log out inactive users after 30 minutes.
NFR-038	The system must implement secure session management.
NFR-039	The system must sanitize user-generated content.
NFR-040	The system must implement audit logging for all critical actions.

5.5 Usability Requirements

ID	Requirement
NFR-041	The system must be intuitive enough for new users to navigate without training.
NFR-042	The system must use consistent design patterns across all pages.
NFR-043	The system must provide clear error messages with resolution steps.
NFR-044	The system must support keyboard navigation for accessibility.
NFR-045	The system must provide tooltips and help icons for features.
NFR-046	The system must include onboarding for new users.
NFR-047	The system must provide search functionality with smart suggestions.
NFR-048	The system must use clear and concise labels.
NFR-049	The system must include a feedback mechanism.
NFR-050	The system must maintain responsive design for all screen sizes.

5.6 Availability Requirements

ID	Requirement
NFR-051	The system must be available 24/7 with a target uptime of 99.9%.
NFR-052	The system must support scheduled maintenance with user notification.
NFR-053	The system must implement disaster recovery procedures.
NFR-054	The system must have automatic failover capability.
NFR-055	The system must provide status monitoring and alerts.

5.7 Maintainability Requirements

ID	Requirement
NFR-056	The system must use modular architecture for easy maintenance.
NFR-057	The system must implement logging for debugging.
NFR-058	The system must support automated testing (unit, integration, e2e).
NFR-059	The system must have clean, documented code.
NFR-060	The system must follow consistent coding standards.
NFR-061	The system must support CI/CD pipeline for deployments.
NFR-062	The system must include health check endpoints.

5.8 Accessibility Requirements

ID	Requirement
NFR-063	The system must comply with WCAG 2.1 Level AA standards.
NFR-064	The system must support screen readers.
NFR-065	The system must provide sufficient color contrast for readability.
NFR-066	The system must support text resizing up to 200%.
NFR-067	The system must include alt text for all images.
NFR-068	The system must support keyboard-only navigation.
NFR-069	The system must provide captions for video content.
NFR-070	The system must support multiple languages (English and Urdu).

5.9 Interoperability Requirements

ID	Requirement
NFR-071	The system must integrate with payment gateways.
NFR-072	The system must integrate with Google Meet for live sessions.
NFR-073	The system must integrate with Zoom for live sessions.
NFR-074	The system must support OAuth with Google and GitHub.
NFR-075	The system must support email delivery services (SendGrid, SES).
NFR-076	The system must integrate with Cloudinary or AWS S3 for media storage.
NFR-077	The system must support RESTful APIs for third-party integration.
NFR-078	The system must support webhooks for external services.

5.10 Compatibility Requirements

ID	Requirement
NFR-079	The system must support Chrome, Firefox, Safari, and Edge (latest two versions).
NFR-080	The system must support mobile browsers (Android Chrome, iOS Safari).
NFR-081	The system must support Windows, macOS, and Linux operating systems.
NFR-082	The system must function on tablets and smartphones.
NFR-083	The system must support Internet Explorer 11 with fallback (basic functionality).
NFR-084	The system must support older browser versions with progressive enhancement.

6. Business Rules

6.1 Membership Rules

ID	Rule
BR-001	All students of UET Lahore and other affiliated institutions are eligible for free membership.
BR-002	External members must pay a nominal registration fee (determined by society).
BR-003	Membership must be renewed annually at the start of the academic year.
BR-004	Members must agree to the society's code of conduct.
BR-005	Members must have a valid university email address for verification.
BR-006	Alumni members have limited access to certain features.
BR-007	Members must complete their profile within 7 days of registration.

6.2 Course Enrollment Rules

ID	Rule
BR-008	Students can enroll in a maximum of 5 courses per semester.
BR-009	Students must complete prerequisite courses before enrolling in advanced courses.
BR-010	Course completion requires passing all quizzes and assignments with 60% minimum score.
BR-011	Paid courses require payment before access.
BR-012	Refunds for paid courses are allowed within 7 days of enrollment (if no lessons accessed).

BR-013	Course certificates are issued only upon full completion.
BR-014	Students must maintain 80% attendance for certificate eligibility.

6.3 Certification Rules

ID	Rule
BR-015	Digital certificates are valid for lifetime unless revoked.
BR-016	Certificates include a unique QR code for verification.
BR-017	Certificate templates must be approved by society admin.
BR-018	Certificates cannot be altered or forged (blockchain/QR verification).
BR-019	Students can request duplicate certificates (with fee).

6.4 Internship Rules

ID	Rule
BR-020	Only registered companies can post internships.
BR-021	Internships must be approved by society admin before publication.
BR-022	Students can apply to a maximum of 5 internships per month.
BR-023	Internship duration must be between 4 weeks and 6 months.
BR-024	Interns must submit weekly progress reports.
BR-025	Internship completion requires mentor evaluation and company feedback.
BR-026	Companies must provide completion certificates for successful interns.

6.5 Event Participation Rules

ID	Rule
BR-027	Registration for events closes 24 hours before the event start.
BR-028	Paid event fees are non-refundable unless event is cancelled.
BR-029	Participants must register with valid contact information.
BR-030	QR codes are valid for single entry only.
BR-031	Participation certificates are issued only to registered attendees.
BR-032	Volunteers must complete training before event day.
BR-033	Organizers must commit to full event duration.

6.6 Community Guidelines

ID	Rule
BR-034	Users must treat all community members with respect.
BR-035	No spam or promotional content in community forums.
BR-036	No harassment, discrimination, or hate speech allowed.
BR-037	Users must report inappropriate content to moderators.
BR-038	Violations may result in temporary or permanent account suspension.
BR-039	Content posted must be original or properly attributed.
BR-040	Repeated violations will result in permanent ban.

7. External Interfaces

7.1 User Interfaces

ID	Interface	Description
UI-001	Landing Page	Public-facing homepage with 3D scene, navigation, and featured content.
UI-002	Authentication Interface	Login, Registration, Password Recovery, and 2FA interfaces.
UI-003	Student Dashboard	Personalized dashboard with courses, events, internships, and progress.
UI-004	Teacher Dashboard	Course management, class creation, grading, and analytics interface.
UI-005	Mentor Dashboard	Internship evaluation, student feedback, and reporting interface.
UI-006	Company Dashboard	Internship posting, application management, and analytics interface.
UI-007	Admin Dashboard	System monitoring, user management, content approval, and analytics.
UI-008	LMS Interface	Course viewing, lesson navigation, assignment submission, and progress tracking.
UI-009	Virtual Classroom	Live session interface with chat, video, and resource sharing.
UI-010	Event Interface	Event browsing, registration, ticketing, and QR attendance.
UI-011	Community Forum	Discussion threads, Q&A, and user interaction interface.
UI-012	Career Portal	Job board, portfolio builder, and career resources.
UI-013	3D Campus	Interactive 3D campus navigation experience.
UI-014	Mobile Responsive Interface	Optimized views for mobile devices.

7.2 Hardware Interfaces

ID	Interface	Description
HW-001	QR Code Scanners	USB/Bluetooth QR scanners for event attendance and certificate verification.
HW-002	Mobile Device Cameras	Camera integration for QR code scanning and profile photos.
HW-003	Computers/Desktops	Primary platform access through web browsers.
HW-004	Tablets	Tablet-optimized interface for event check-in and administration.
HW-005	Printers	Certificate printing (optional physical copies).

7.3 Software Interfaces

ID	Interface	Description
SI-001	Payment Gateway	Integration with JazzCash, EasyPaisa, and bank APIs for payments.
SI-002	Email Service	Integration with SendGrid/AWS SES for email notifications.
SI-003	SMS Gateway	Integration for SMS notifications (optional).
SI-004	Google Meet API	Integration for live session creation and management.
SI-005	Zoom API	Integration for Zoom live sessions.
SI-006	Cloud Storage	Integration with Cloudinary/S3 for media storage.
SI-007	OAuth Providers	Google and GitHub OAuth integration for authentication.
SI-008	AI APIs	Integration with OpenAI/Claude for AI features.
SI-009	Analytics Services	Integration with Google Analytics/Plausible for tracking.
SI-010	Security Services	Integration with security scanning and monitoring services.

7.4 Communication Interfaces

ID	Interface	Description
CI-001	HTTPS	All communication encrypted with TLS 1.3.
CI-002	REST API	Secure RESTful APIs for client-server communication.
CI-003	WebSocket	Real-time communication for notifications and live sessions.
CI-004	WebRTC	Peer-to-peer audio/video for live sessions.
CI-005	Email	SMTP-based email delivery for notifications.
CI-006	SMS	SMS delivery for critical notifications (optional).
CI-007	Push Notifications	Web push notifications for real-time alerts.

8. System Architecture

8.1 Architecture Overview

The **Inquisitors Society Platform** follows a modern, scalable architecture based on microservices and client-server design. The architecture is designed to be:

- **Scalable:** Supports horizontal scaling for all components
- **Maintainable:** Modular design with clear separation of concerns
- **Secure:** Comprehensive security from client to database
- **High-Performance:** Optimized for fast response times
- **Resilient:** Built with failure recovery mechanisms

8.2 Technology Stack

8.2.1 Frontend Technology Stack

Technology	Version	Purpose
Next.js	14+	React framework with SSR, SSG, and routing
React	18+	UI component library
TypeScript	5+	Type-safe JavaScript
Tailwind CSS	3+	Utility-first CSS framework
Three.js	0.160+	3D rendering engine
GSAP	3.12+	Professional animations
Framer Motion	10+	React animation library
Spline	-	3D design tool integration
Zustand	4+	State management
React Query	5+	Server state management
Socket.IO Client	4+	Real-time communication
Axios	1+	HTTP client

8.2.2 Backend Technology Stack

Technology	Version	Purpose
NestJS	10+	Backend framework
Node.js	20+	Runtime environment
PostgreSQL	15+	Relational database
Redis	7+	Caching and session management
Prisma ORM	5+	Database ORM
Socket.IO	4+	Real-time communication
JWT	9+	Authentication
Bcrypt	5+	Password hashing
Multer	2+	File uploads
Class-Validator	0.14+	Input validation

8.2.3 Infrastructure Stack

Technology	Version	Purpose
Docker	24+	Containerization
Docker Compose	2.23+	Multi-container orchestration
Kubernetes	1.28+	Container orchestration
AWS/GCP/Azure	-	Cloud platform
Nginx	1.24+	Reverse proxy and load balancing
PostgreSQL Replication	15+	Database replication
Redis Cluster	7+	Distributed caching
Cloudinary	-	Media storage and optimization
AWS S3	-	File storage backup

8.3 Frontend Architecture

8.3.1 Architecture Pattern

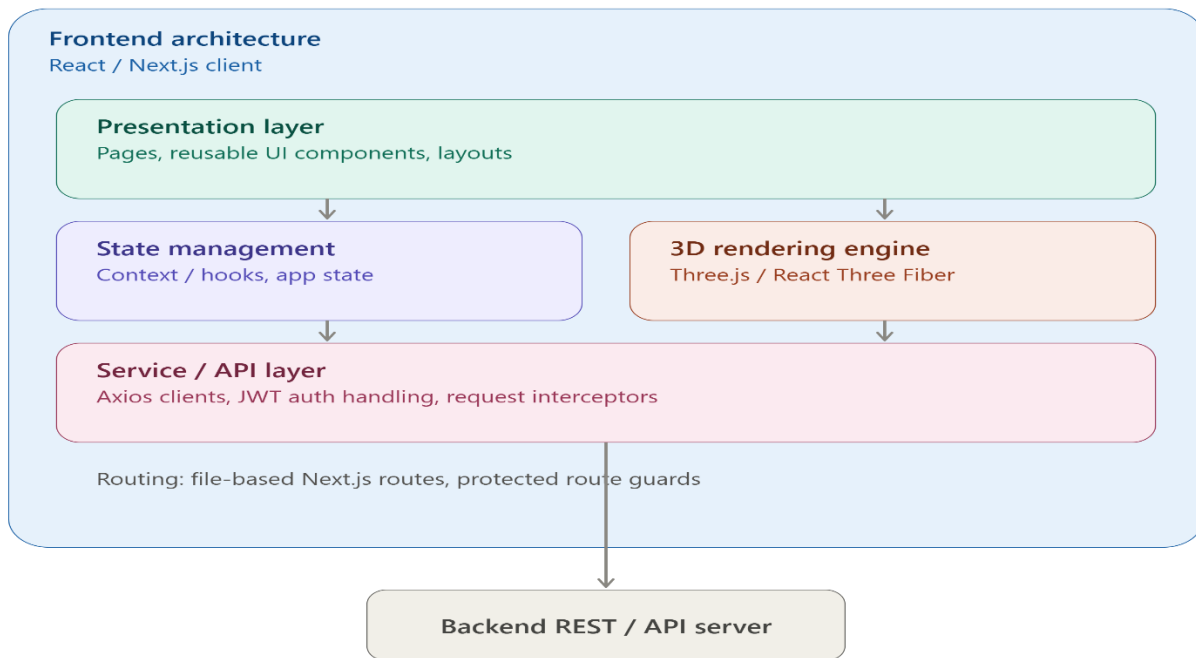


Figure 1: Frontend Architecture

8.4 Backend Architecture

8.4.1 Architecture Pattern

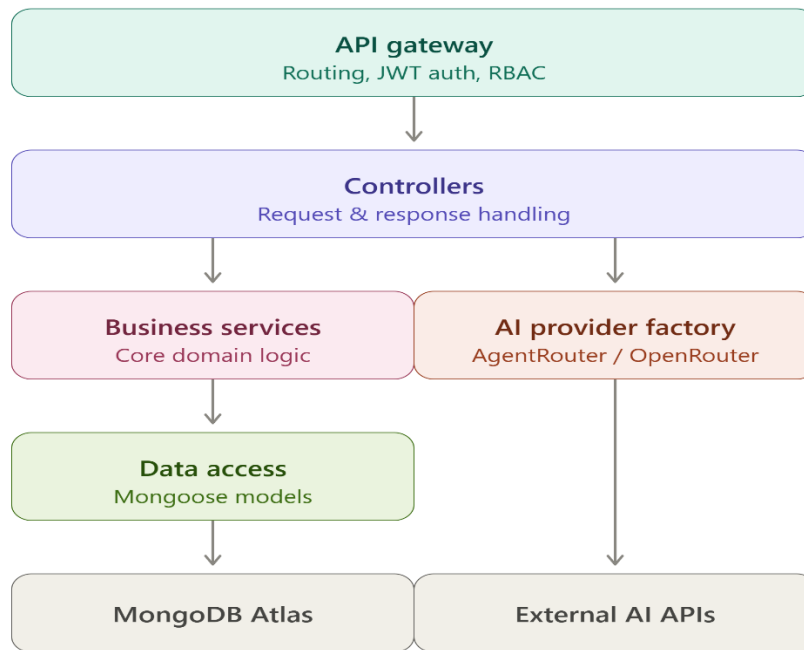


Figure 2: Backend Architecture

8.5 Database Architecture

8.5.1 Database Design Principles

- **Normalization:** 3NF normalization for efficient data storage
- **Indexing:** Proper indexes on frequently queried columns
- **Relationships:** Clear foreign key relationships
- **Constraints:** Data integrity through constraints
- **Auditing:** Audit tables for critical data changes

8.5.2 Core Entities

1. **User:** All platform users (students, teachers, mentors, companies, admins)
2. **Profile:** Extended user information
3. **Role:** User roles and permissions
4. **Course:** Course information and metadata
5. **Module:** Course modules/sections
6. **Lesson:** Individual lessons within modules

7. **Assignment:** Assignments and submissions
8. **Quiz:** Quizzes and questions
9. **Enrollment:** Student course enrollments
10. **Class:** Virtual classrooms
11. **Internship:** Internship postings
12. **Application:** Student internship applications
13. **Event:** Event information
14. **Registration:** Event registrations
15. **Certificate:** Digital certificates
16. **ForumTopic:** Community discussion topics
17. **Comment:** Comments on forums
18. **Notification:** User notifications
19. **AuditLog:** System audit trail
20. **Settings:** Platform configuration

8.6 3D Rendering Architecture

8.6.1 Rendering Pipeline

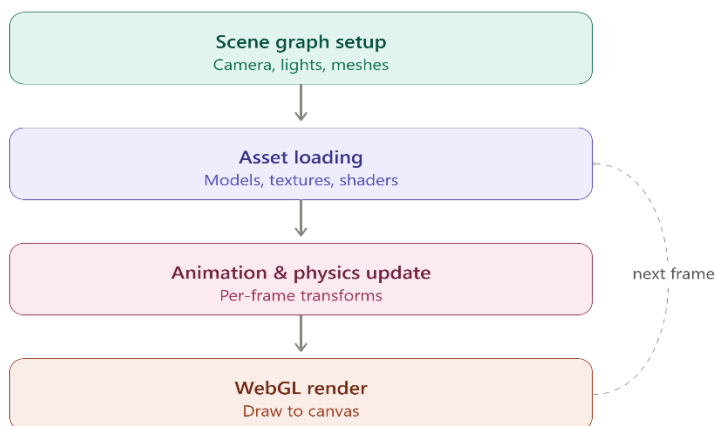


Figure 3: Rendering Pipeline

8.7 AI Integration Architecture

8.7.1 AI Services Integration

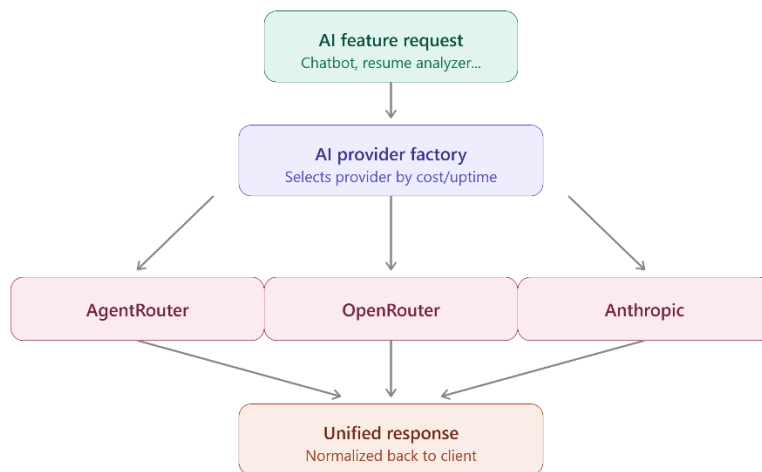


Figure 4: AI Services Integration

8.8 Security Architecture

8.8.1 Security Layers

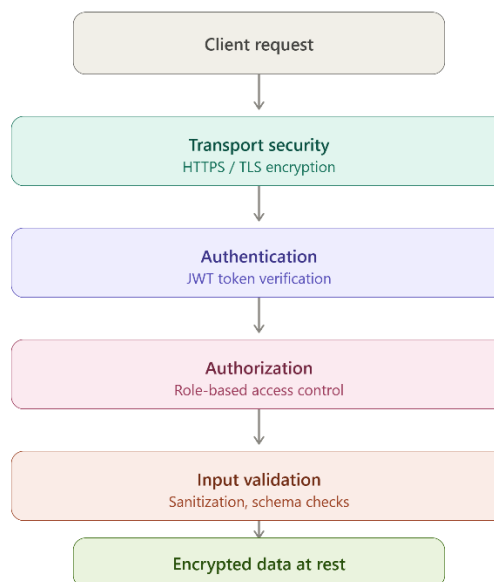


Figure 5: Security Architecture

9. Database Design

9.1 Entity-Relationship Diagram

9.2 Entity Definitions

9.2.1 Users Table

Field	Type	Description
id	UUID	Primary key
email	VARCHAR(255)	Unique email address
password_hash	VARCHAR(255)	Bcrypt hash of password
full_name	VARCHAR(255)	User's full name
cnic	VARCHAR(15)	National identity card number
phone	VARCHAR(20)	Phone number
role	ENUM	student, teacher, mentor, company, admin
is_verified	BOOLEAN	Email verification status
is_active	BOOLEAN	Account active status
last_login	TIMESTAMP	Last login timestamp
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp
deleted_at	TIMESTAMP	Soft delete timestamp

9.2.2 Profiles Table

Field	Type	Description
id	UUID	Primary key
user_id	UUID	Foreign key to users
avatar_url	TEXT	Profile picture URL
bio	TEXT	User biography
university	VARCHAR(255)	University name
department	VARCHAR(255)	Department
year	VARCHAR(50)	Year of study
skills	JSONB	Array of skills
interests	JSONB	Array of interests
social_links	JSONB	Social media links
github_url	TEXT	GitHub URL
linkedin_url	TEXT	LinkedIn URL
portfolio_url	TEXT	Portfolio URL
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.3 Courses Table

Field	Type	Description
-------	------	-------------

id	UUID	Primary key
title	VARCHAR(255)	Course title
slug	VARCHAR(255)	URL-friendly slug
description	TEXT	Course description
category_id	UUID	Foreign key to categories
instructor_id	UUID	Foreign key to users (teachers)
thumbnail_url	TEXT	Course thumbnail
price	DECIMAL(10,2)	Course price (0 for free)
level	ENUM	beginner, intermediate, advanced
duration_hours	INTEGER	Estimated duration in hours
learning_objectives	JSONB	Array of learning objectives
prerequisites	JSONB	Array of prerequisites
is_published	BOOLEAN	Publication status
is_featured	BOOLEAN	Featured course status
rating	DECIMAL(3,2)	Average rating
students_count	INTEGER	Number of enrolled students
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.4 Modules Table

Field	Type	Description
id	UUID	Primary key
course_id	UUID	Foreign key to courses
title	VARCHAR(255)	Module title
description	TEXT	Module description
order_index	INTEGER	Module order in course
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.5 Lessons Table

Field	Type	Description
id	UUID	Primary key
module_id	UUID	Foreign key to modules
title	VARCHAR(255)	Lesson title
lesson_type	ENUM	video, text, assignment, quiz, lab, presentation
content	JSONB	Lesson content (depending on type)
video_url	TEXT	Video URL (if applicable)
resources	JSONB	Array of resource URLs
duration_minutes	INTEGER	Estimated duration in minutes
is_free	BOOLEAN	Free lesson status
order_index	INTEGER	Lesson order in module
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.6 Enrollments Table

Field	Type	Description
id	UUID	Primary key
user_id	UUID	Foreign key to users (student)
course_id	UUID	Foreign key to courses
progress	FLOAT	Course progress (0-100)
completed_at	TIMESTAMP	Completion timestamp
certificate_id	UUID	Foreign key to certificates
is_active	BOOLEAN	Enrollment active status
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.7 Assignments Table

Field	Type	Description
id	UUID	Primary key
lesson_id	UUID	Foreign key to lessons
title	VARCHAR(255)	Assignment title
instructions	TEXT	Assignment instructions
deadline	TIMESTAMP	Submission deadline
max_score	INTEGER	Maximum possible score
rubric	JSONB	Grading rubric
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.8 Submissions Table

Field	Type	Description
id	UUID	Primary key
assignment_id	UUID	Foreign key to assignments
user_id	UUID	Foreign key to users (student)
submission_text	TEXT	Text submission
submission_files	JSONB	Array of file URLs
submitted_at	TIMESTAMP	Submission timestamp
grade	FLOAT	Assignment grade
feedback	TEXT	Teacher feedback
is_grade_published	BOOLEAN	Grade visibility status
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.9 Quizzes Table

Field	Type	Description
id	UUID	Primary key
lesson_id	UUID	Foreign key to lessons

title	VARCHAR(255)	Quiz title
description	TEXT	Quiz description
time_limit_minutes	INTEGER	Time limit in minutes
passing_score	INTEGER	Passing score percentage
is_practice	BOOLEAN	Practice quiz flag
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.10 Questions Table

Field	Type	Description
id	UUID	Primary key
quiz_id	UUID	Foreign key to quizzes
question_text	TEXT	Question text
question_type	ENUM	mcq, true_false, fill_blank, matching, essay
options	JSONB	Question options (for MCQs)
correct_answer	TEXT	Correct answer (for objective questions)
points	INTEGER	Points for question
order_index	INTEGER	Question order
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.11 Certificates Table

Field	Type	Description
id	UUID	Primary key
user_id	UUID	Foreign key to users
course_id	UUID	Foreign key to courses (optional)
event_id	UUID	Foreign key to events (optional)
internship_id	UUID	Foreign key to internships (optional)
certificate_code	VARCHAR(50)	Unique certificate code
verification_code	VARCHAR(50)	QR verification code
issue_date	DATE	Certificate issue date
expiry_date	DATE	Certificate expiry date
certificate_url	TEXT	Download URL
status	ENUM	active, revoked, expired
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.12 Internships Table

Field	Type	Description
id	UUID	Primary key
company_id	UUID	Foreign key to users (company)
title	VARCHAR(255)	Internship title
description	TEXT	Detailed description

requirements	JSONB	Skills and requirements
duration	VARCHAR(50)	Duration string
stipend	VARCHAR(50)	Stipend information
location	VARCHAR(255)	Location
positions	INTEGER	Number of positions
application_deadline	TIMESTAMP	Deadline
type	ENUM	paid, unpaid, credit
status	ENUM	draft, pending, open, closed
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.13 Applications Table

Field	Type	Description
id	UUID	Primary key
internship_id	UUID	Foreign key to internships
user_id	UUID	Foreign key to users (student)
cover_letter	TEXT	Cover letter
cv_url	TEXT	CV upload URL
status	ENUM	submitted, reviewed, shortlisted, accepted, rejected
mentor_id	UUID	Foreign key to users (mentor)
mentor_feedback	TEXT	Mentor feedback
mentor_score	INTEGER	Mentor evaluation score
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.14 Events Table

Field	Type	Description
id	UUID	Primary key
title	VARCHAR(255)	Event title
description	TEXT	Event description
type	ENUM	seminar, workshop, hackathon, conference, competition, tour, book_fair, leadership, css, ielts
category	VARCHAR(100)	Event category
venue	VARCHAR(255)	Location/Venue
start_date	TIMESTAMP	Event start
end_date	TIMESTAMP	Event end
registration_deadline	TIMESTAMP	Registration deadline
capacity	INTEGER	Maximum participants
price	DECIMAL(10,2)	Registration fee
is_paid	BOOLEAN	Paid event flag
status	ENUM	draft, pending, approved, published, ongoing, completed, cancelled
organizer_id	UUID	Foreign key to users
speakers	JSONB	Array of speakers

agenda	JSONB	Event schedule
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.15 Registrations Table

Field	Type	Description
id	UUID	Primary key
event_id	UUID	Foreign key to events
user_id	UUID	Foreign key to users
registration_type	ENUM	participant, volunteer, organizer
payment_status	ENUM	pending, paid, refunded
payment_id	UUID	Foreign key to payments
qr_code	TEXT	QR code data
attendance_status	BOOLEAN	Attended or not
certificate_id	UUID	Foreign key to certificates
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.16 Discussion Threads Table

Field	Type	Description
id	UUID	Primary key
title	VARCHAR(255)	Thread title
content	TEXT	Thread content
category	ENUM	general, technical, career, events, research, leadership
author_id	UUID	Foreign key to users
is_pinned	BOOLEAN	Pinned status
is_locked	BOOLEAN	Locked status
views	INTEGER	View count
replies_count	INTEGER	Reply count
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.2.17 Comments Table

Field	Type	Description
id	UUID	Primary key
thread_id	UUID	Foreign key to discussion threads
user_id	UUID	Foreign key to users
content	TEXT	Comment content
parent_comment_id	UUID	Foreign key to parent comment
likes	INTEGER	Like count
is_best_answer	BOOLEAN	Best answer flag
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

9.3 Relationships

Relationship	Type	Description
User -> Profile	One-to-One	Each user has one profile
User -> Enrollment	One-to-Many	A user can enroll in many courses
Course -> Module	One-to-Many	Each course has many modules
Module -> Lesson	One-to-Many	Each module has many lessons
Lesson -> Assignment	One-to-One	Each lesson can have one assignment
Lesson -> Quiz	One-to-One	Each lesson can have one quiz
User -> Submission	One-to-Many	A user can make many submissions
Assignment -> Submission	One-to-Many	An assignment can have many submissions
Course -> Certificate	One-to-Many	A course can have many certificates
User -> Certificate	One-to-Many	A user can have many certificates
Company -> Internship	One-to-Many	A company can post many internships
Internship -> Application	One-to-Many	An internship can have many applications
User -> Application	One-to-Many	A user can have many applications
Event -> Registration	One-to-Many	An event can have many registrations
User -> Registration	One-to-Many	A user can have many registrations
Thread -> Comment	One-to-Many	A thread can have many comments
Comment -> Comment	One-to-Many	A comment can have many replies

9.4 Indexing Strategy

Table	Index	Type
users	email	UNIQUE
users	role	B-Tree
courses	slug	UNIQUE
courses	category_id	B-Tree
courses	instructor_id	B-Tree
courses	is_published	B-Tree
enrollments	user_id, course_id	Composite
enrollments	course_id	B-Tree
lessons	module_id	B-Tree
assignments	lesson_id	B-Tree
submissions	assignment_id	B-Tree
submissions	user_id	B-Tree
internships	company_id	B-Tree
internships	status	B-Tree
applications	internship_id	B-Tree
applications	user_id	B-Tree
applications	status	B-Tree
events	start_date	B-Tree
events	status	B-Tree
registrations	event_id	B-Tree

registrations	user_id	B-Tree
certificates	certificate_code	UNIQUE
certificates	verification_code	UNIQUE
discussion_threads	category	B-Tree
discussion_threads	author_id	B-Tree
comments	thread_id	B-Tree

10. Use Cases

10.1 Use Case Summary Table

Use Case ID	Name	Actor
UC-01-001	Register Account	Student, Teacher, Mentor, Company
UC-01-002	Login to System	All Users
UC-01-003	Manage Profile	All Users
UC-01-004	Browse Courses	Student
UC-01-005	Enroll in Course	Student
UC-01-006	View Course Content	Student
UC-01-007	Submit Assignment	Student
UC-01-008	Take Quiz	Student
UC-01-009	Track Progress	Student
UC-01-010	Download Certificate	Student
UC-02-001	Create Course	Teacher
UC-02-002	Create Lesson	Teacher
UC-02-003	Create Assignment	Teacher
UC-02-004	Create Quiz	Teacher
UC-02-005	Grade Assignment	Teacher
UC-02-006	View Student Progress	Teacher
UC-02-007	Create Virtual Class	Teacher
UC-02-008	Conduct Live Session	Teacher
UC-03-001	Post Internship	Company
UC-03-002	Review Applications	Company
UC-03-003	Evaluate Internship	Mentor
UC-03-004	Provide Feedback	Mentor
UC-04-001	Create Event	Admin/Organizer
UC-04-002	Manage Registrations	Admin/Organizer
UC-04-003	Scan QR Attendance	Admin/Organizer
UC-05-001	Manage Users	Admin
UC-05-002	Approve Content	Admin
UC-05-003	View Analytics	Admin
UC-05-004	Manage System Settings	Admin

10.2 Detailed Use Cases

10.2.1 Student Use Cases

UC-01-001: Register Account

Element	Description
ID	UC-01-001
Name	Register Account
Actor	Student
Goal	Create a new student account on the platform
Trigger	Student clicks "Sign Up" on the landing page
Pre-conditions	Student has valid email address and internet connection
Post-conditions	Account created; verification email sent
Main Flow	1. Student navigates to registration page 2. Enters personal information (name, email, phone, CNIC) 3. Selects "Student" as role 4. Enters university, department, year of study 5. Creates password 6. Submits registration form 7. System validates information 8. System creates account and sends verification email 9. System displays success message
Alternate Flow	University email domain validation fails System prompts student to use valid university email
Exception	Email already registered System shows error and suggests password recovery
Quality	Registration completes within 30 seconds

UC-01-005: Enroll in Course

Element	Description
ID	UC-01-005
Name	Enroll in Course
Actor	Student
Goal	Enroll in a course on the platform
Trigger	Student clicks "Enroll" on course detail page
Pre-conditions	Student is logged in; course is available
Post-conditions	Student enrolled; course appears in dashboard
Main Flow	1. Student browses course catalog 2. Clicks on course to view details 3. Reviews course information 4. Clicks "Enroll" button 5. If course is free, system confirms enrollment 6. If course is paid, system redirects to payment 7. System adds student to course

	8. System displays success message 9. System sends enrollment confirmation email
Alternate Flow	Paid course: Student completes payment process
Exception	Course is full; System shows waitlist option
Quality	Enrollment completes within 5 seconds

10.2.2 Teacher Use Cases

UC-02-001: Create Course

Element	Description
ID	UC-02-001
Name	Create Course
Actor	Teacher
Goal	Create a new course on the platform
Trigger	Teacher clicks "Create Course" on dashboard
Pre-conditions	Teacher is logged in and verified
Post-conditions	Course created as draft; pending approval
Main Flow	1. Teacher clicks "Create Course" 2. Enters course details (title, description, category) 3. Sets learning objectives and prerequisites 4. Selects course level (beginner, intermediate, advanced) 5. Sets price (free or paid) 6. Uploads course thumbnail 7. Clicks "Save as Draft" 8. System creates course and saves metadata 9. System displays success message
Alternate Flow	Teacher clicks "Publish" - system submits for approval
Exception	Required fields missing; system shows validation errors
Quality	Course creation completes within 2 minutes

10.2.3 Mentor Use Cases

UC-03-003: Evaluate Internship

Element	Description
ID	UC-03-003
Name	Evaluate Internship Application
Actor	Mentor
Goal	Evaluate student applications for internships
Trigger	Mentor receives notification of new application
Pre-conditions	Mentor is logged in; application is pending review
Post-conditions	Application status updated; feedback recorded

Main Flow	1. Mentor logs in to dashboard 2. Views pending internship applications 3. Selects an application to review 4. Reviews student profile and attached documents 5. Provides evaluation score and comments 6. Selects status (shortlist/reject) 7. Submits evaluation 8. System updates application status 9. System notifies student of decision
Exception	Student profile incomplete; mentor requests additional information
Quality	Evaluation form loads within 2 seconds

10.2.4 Company Use Cases

UC-03-001: Post Internship

Element	Description
ID	UC-03-001
Name	Post Internship
Actor	Company
Goal	Post a new internship opportunity
Trigger	Company clicks "Post Internship" on dashboard
Pre-conditions	Company is registered and verified
Post-conditions	Internship posted; pending admin approval
Main Flow	1. Company logs in to dashboard 2. Clicks "Post New Internship" 3. Enters internship details (title, description, requirements) 4. Sets duration, stipend, location, positions 5. Selects internship type (paid/unpaid/credit) 6. Sets application deadline 7. Reviews and submits 8. System creates internship 9. System submits for admin approval 10. System displays success message
Exception	Details incomplete; system shows validation errors
Quality	Posting submission completes within 3 minutes

10.2.5 Admin Use Cases

UC-05-001: Manage Users

Element	Description
ID	UC-05-001
Name	Manage Users
Actor	Admin
Goal	Manage platform users (view, edit, delete)

Trigger	Admin clicks "User Management" on dashboard
Pre-conditions	Admin is logged in with appropriate permissions
Post-conditions	User management actions completed
Main Flow	1. Admin opens User Management 2. Views list of all users with filters 3. Searches for specific user 4. Selects user to view details 5. Performs action (edit role, activate, deactivate, delete) 6. Confirms action 7. System applies changes 8. System logs action in audit
Exception	Cannot delete user with active enrollments; system shows warning
Quality	User list loads within 2 seconds

11. Acceptance Criteria

11.1 Functional Acceptance Criteria

ID	Criteria
AC-001	All user registration flows work correctly for all roles (Student, Teacher, Mentor, Company, Admin).
AC-002	Authentication works with email/password and OAuth (Google, GitHub).
AC-003	Role-based access control correctly restricts features based on user roles.
AC-004	Users can create, edit, and delete profiles successfully.
AC-005	Teachers can create courses with all required fields.
AC-006	Teachers can create lessons of all types (video, text, assignment, quiz, lab, presentation).
AC-007	Students can enroll in courses and access content.
AC-008	Students can submit assignments and view grades.
AC-009	Students can take quizzes and view results.
AC-010	Teachers can grade assignments and provide feedback.
AC-011	Companies can post internships with all required details.
AC-012	Students can apply for internships and track application status.
AC-013	Mentors can evaluate internship applications and provide feedback.
AC-014	Admins can create events and manage registrations.
AC-015	QR code ticketing works for event attendance.
AC-016	Certificate generation works for courses, events, and internships.
AC-017	Certificate verification works with QR codes.
AC-018	AI chatbot provides relevant responses to user queries.
AC-019	AI resume analyzer provides meaningful feedback.
AC-020	AI career advisor provides relevant recommendations.
AC-021	3D landing scene loads and displays correctly.
AC-022	3D campus navigation works with interactive buildings.

AC-023	Community forums support creating threads and comments.
AC-024	Notification system sends notifications for all relevant events.
AC-025	Admin dashboard displays accurate analytics.

11.2 Non-Functional Acceptance Criteria

ID	Criteria
AC-026	System responds to user interactions within 2 seconds (95th percentile).
AC-027	Landing page loads within 3 seconds on standard broadband.
AC-028	System supports 10,000 concurrent users without performance degradation.
AC-029	System maintains 99.9% uptime during operational hours.
AC-030	All data at rest is encrypted with AES-256.
AC-031	All data in transit is encrypted with TLS 1.3.
AC-032	Passwords are hashed using bcrypt.
AC-033	System passes OWASP Top 10 security checks.
AC-034	System achieves WCAG 2.1 Level AA accessibility compliance.
AC-035	System supports Chrome, Firefox, Safari, and Edge (latest two versions).
AC-036	System is responsive for mobile, tablet, and desktop.
AC-037	Database query response time is under 100ms for common operations.
AC-038	3D scenes load within 5 seconds with caching.

11.3 UI/UX Acceptance Criteria

ID	Criteria
AC-039	Design follows modern UI principles with consistent branding.
AC-040	Navigation is intuitive and requires no training for basic tasks.
AC-041	Color contrast meets WCAG AA standards.
AC-042	Font sizes are readable on all devices.
AC-043	All forms have clear labels and validation errors.
AC-044	Loading states and progress indicators are displayed.
AC-045	Mobile interface is touch-friendly with adequate tap targets.
AC-046	3D scenes enhance user experience without being distracting.
AC-047	Animations are smooth and improve usability.
AC-048	Content is available in both English and Urdu.

12. Risk Analysis

12.1 Technical Risks

Risk ID	Description	Likelihood	Impact	Mitigation
---------	-------------	------------	--------	------------

R-001	3D rendering may cause performance issues on low-end devices	Medium	High	Implement LOD, asset optimization, fallback for low-end devices
R-002	AI API costs may exceed budget	Medium	Medium	Implement caching, rate limiting, use cheaper models
R-003	Database scalability issues with growth	Low	High	Use read replicas, sharding strategy, optimize queries
R-004	Third-party API integration failures	Medium	Medium	Implement fallbacks, retry mechanisms, monitoring
R-005	Security vulnerabilities in dependencies	Low	High	Regular dependency updates, security scanning
R-006	Live session quality issues (Google Meet/Zoom)	Low	Medium	Provide alternative connectivity options
R-007	File storage limitations	Low	Medium	Implement file size limits, compression, archival

12.2 Operational Risks

Risk ID	Description	Likelihood	Impact	Mitigation
R-008	User adoption is lower than expected	Medium	Medium	Marketing campaign, user onboarding, incentives
R-009	Content quality issues	Medium	Medium	Content review process, instructor training
R-010	Staff availability for content creation	Medium	Medium	Recruit additional instructors, automate
R-011	Data privacy compliance issues	Low	High	GDPR compliance, data protection policy
R-012	Server downtime during peak usage	Medium	High	Auto-scaling, monitoring, load testing

12.3 Security Risks

Risk ID	Description	Likelihood	Impact	Mitigation
R-013	Data breach	Low	High	Encryption, security audits, monitoring
R-014	Unauthorized access	Low	High	RBAC, MFA, regular audits
R-015	DDoS attacks	Medium	High	DDoS protection, rate limiting
R-016	API abuse	Medium	Medium	Rate limiting, API keys, monitoring

12.4 Mitigation Strategies

12.4.1 Performance Mitigation

- Implement caching layers (Redis, CDN)
- Use lazy loading and code splitting
- Optimize images and assets
- Monitor performance metrics
- Conduct regular load testing

12.4.2 Security Mitigation

- Regular security audits
- Penetration testing
- Dependency vulnerability scanning
- Incident response plan
- Regular backup and recovery testing

12.4.3 Operational Mitigation

- Comprehensive monitoring and alerting
- Disaster recovery plan
- Regular staff training
- Content review process
- User feedback system

13. Future Scope

13.1 Phase 2 Features

Feature	Description	Priority
Mobile Native Apps	iOS and Android native applications	High
Gamification	Points, badges, leaderboards	Medium
Peer-to-Peer Mentorship	Student-to-student mentorship matching	Medium
Virtual Reality (VR) Campus	VR experience for campus navigation	Low
Blockchain Certificates	Certificate verification on blockchain	Low
Multi-Institution Support	Support for multiple universities	Medium
Corporate Training	Corporate training packages	Medium
Advanced Analytics	Predictive analytics and insights	Medium
SSO Integration	Single sign-on with university systems	Medium
Offline Capability	Offline learning and syncing	Medium

13.2 Phase 3 Features

Feature	Description	Priority
AI-Powered Grading	Automatic assignment grading	High
Adaptive Learning	Personalized learning paths	Medium
Language Translation	Real-time content translation	Low
Video Transcription	AI-powered video transcription	Medium

Speech-to-Text	Voice commands and dictation	Low
AR Features	Augmented reality for events	Low
Student Marketplace	Exchange of notes and resources	Low
Alumni Network	Alumni engagement platform	Medium
Scholarship Management	Scholarship applications and management	Low
Conference Management	End-to-end conference management	Medium

14. Development Roadmap

14.1 Phase 0: Requirements & Planning

Activity	Duration	Deliverables	Tools
Requirement Gathering	2 weeks	Requirements document, User stories	Interviews, Surveys, Workshops
Business Analysis	1 week	Business process documentation, BRD	BPMN tools
Research	1 week	Technology selection, Architecture decision	Research reports
Software Planning	1 week	Project plan, Timeline, Resource allocation	Jira, Gantt charts
Sprint Planning	3 days	Sprint backlog, Story points	Jira, Agile methodology

Team:

- Project Manager
- Business Analyst
- Senior Developer

Timeline: 5 weeks

Milestone: Requirements approved

14.2 Phase 1: UI/UX & 3D Design

Activity	Duration	Deliverables	Tools
Wireframes	2 weeks	Low-fidelity wireframes for all pages	Figma, Sketch
High-Fidelity UI Design	4 weeks	Full UI design with branding	Figma, Adobe XD
Design System	2 weeks	Component library, Design tokens	Figma
3D Scene Design	4 weeks	Campus layout, 3D models, Animations	Spline, Blender, Three.js
UI Prototype	2 weeks	Clickable prototype	Figma, InVision

Team:

- UI/UX Designer (2)
- 3D Designer (1)
- Creative Director

Timeline: 8 weeks

Milestone: Design approved

UI/UX Process:

1. **User Research:** Understand target users
2. **Information Architecture:** Structure content
3. **Wireframing:** Low-fidelity layouts
4. **Prototyping:** Interactive mockups
5. **Visual Design:** High-fidelity designs
6. **Usability Testing:** Test with real users
7. **Iteration:** Refine based on feedback

3D Design Process:

1. **Concept Development:** Define 3D experience
2. **Scene Design:** Layout and composition
3. **Modeling:** Create 3D models
4. **Animation:** Add movement and interactivity
5. **Optimization:** Ensure performance
6. **Integration:** Add to application

Recommended 3D Experience:

- **Knowledge Cube:** A rotating cube with floating educational icons representing different modules. This creates immediate engagement and communicates the platform's comprehensive nature.
- **Interactive Campus:** Buildings represent different modules (LMS Building, Internship Tower, Event Hall, Career Center, Community Hub). Users can hover and click to navigate.
- **Career Timeline:** A 3D timeline/path with milestones representing career growth stages. This visually communicates career development.
- **Floating Books:** Animated books floating around the platform, symbolizing knowledge and learning.
- **Particle System:** Dynamic particles in the background creating a futuristic, intelligent atmosphere.
- **Achievement Wall:** A 3D wall displaying certificates, medals, and achievements with reveal animations.
- **Virtual Auditorium:** A 3D auditorium space for events and conferences.

Why Each Animation:

- **Parallax:** Creates depth and immersion, making the experience feel premium

- **Smooth Camera:** Professional feel, guides user attention
- **Particle System:** Creates energetic, intelligent atmosphere
- **Glassmorphism:** Modern, clean aesthetic that feels current
- **Reveal Animations:** Builds excitement and guides attention to important content

14.3 Phase 2: Database & Backend Development

Activity	Duration	Deliverables	Tools
Database Design	2 weeks	ERD, Data dictionary, Migration scripts	PostgreSQL, Prisma
Backend Architecture	1 week	Architecture diagrams, API design	NestJS, TypeScript
Authentication Service	3 weeks	JWT, OAuth, MFA implementation	NestJS, JWT, OAuth
User Service	2 weeks	User management, Profiles	NestJS, PostgreSQL
LMS Service	4 weeks	Courses, Lessons, Assignments, Quizzes	NestJS, PostgreSQL
Virtual Class Service	3 weeks	Google Meet/Zoom integration, Live sessions	NestJS, Google Meet API
Internship Service	3 weeks	Internship posting, Applications, Workflow	NestJS, PostgreSQL
Event Service	2 weeks	Event management, QR ticketing	NestJS, PostgreSQL
Community Service	2 weeks	Forums, Q&A, Notifications	NestJS, Socket.IO
Career Service	2 weeks	Portfolio, Job board	NestJS, PostgreSQL
AI Service Integration	3 weeks	Chatbot, Resume analyzer, Career advisor	OpenAI API, NestJS
Payment Integration	2 weeks	Payment gateway integration	JazzCash, EasyPaisa APIs
Notification Service	2 weeks	Email, Push notifications	SendGrid, Firebase
Admin Panel	3 weeks	Admin dashboard, Analytics	NestJS, PostgreSQL

Team:

- Backend Lead (1)
- Backend Developers (3-4)
- Database Administrator (1)
- DevOps Engineer (1)

Timeline: 12 weeks

Milestone: Backend MVP complete

Database Development Guidelines:

1. **Use Prisma for migrations:** Ensures version control of schema
2. **Implement database indexes:** For performance optimization
3. **Use UUID as primary keys:** For distributed systems
4. **Implement soft delete:** For data retention
5. **Use JSONB for flexible data:** For dynamic fields

6. Implement row-level security: For multi-tenant isolation

14.4 Phase 3: Frontend Development

Activity	Duration	Deliverables	Tools
Frontend Architecture	1 week	Project structure, State management	Next.js, Zustand, React Query
Layout & Navigation	2 weeks	Main layout, Navigation, Routing	Next.js, Tailwind CSS
Auth Pages	2 weeks	Login, Register, Forgot password	Next.js, Tailwind CSS
Student Dashboard	3 weeks	Dashboard, Profile, Courses	Next.js, Tailwind CSS
Teacher Dashboard	3 weeks	Dashboard, Course management	Next.js, Tailwind CSS
Mentor Dashboard	2 weeks	Dashboard, Evaluation interface	Next.js, Tailwind CSS
Company Dashboard	2 weeks	Dashboard, Internship management	Next.js, Tailwind CSS
Admin Dashboard	4 weeks	Dashboard, User management, Analytics	Next.js, Tailwind CSS
LMS Frontend	4 weeks	Course viewing, Progress tracking	Next.js, Tailwind CSS
Virtual Classroom	3 weeks	Live session interface	Next.js, WebRTC
Event Pages	3 weeks	Event browsing, Registration, Ticketing	Next.js, Tailwind CSS
Community Forum	3 weeks	Forums, Q&A, Comments	Next.js, Tailwind CSS
Career Portal	3 weeks	Job board, Portfolio, Resume builder	Next.js, Tailwind CSS
3D Integration	4 weeks	Landing scene, Interactive campus	Three.js, GSAP, Framer Motion
Mobile Responsive	3 weeks	Mobile-optimized views	Tailwind CSS
Accessibility	2 weeks	WCAG compliance	ARIA, Accessibility tools

Team:

- Frontend Lead (1)
- Frontend Developers (3-4)
- 3D Developer (1)

Timeline: 14 weeks

Milestone: Frontend complete

Frontend Development Guidelines:

1. **Mobile-first development:** Start with mobile designs
2. **Component-driven development:** Reusable components
3. **TypeScript for type safety:** Catch bugs early
4. **State management:** Use Zustand + React Query
5. **Performance optimization:** Code splitting, lazy loading
6. **Accessibility first:** WCAG 2.1 AA compliance

14.5 Phase 4: AI Module Development

Activity	Duration	Deliverables	Tools
AI Chatbot	4 weeks	Conversational AI assistant	OpenAI API, LangChain
Resume Analyzer	3 weeks	Resume analysis and feedback	OpenAI API, NLP
Career Advisor	3 weeks	Career recommendations	OpenAI API, ML
Skill Gap Analysis	3 weeks	Skill assessment and gap detection	OpenAI API, ML
Interview Practice	3 weeks	Mock interview simulation	OpenAI API, Speech-to-Text
AI Quiz Generator	2 weeks	Automatic quiz generation	OpenAI API, NLP
AI Integration Testing	2 weeks	Integration with main platform	Jest, Testing Library

Team:

- AI Engineer (1-2)
- Backend Developer (1)

Timeline: 8 weeks

Milestone: AI features complete

14.6 Phase 5: Integration & Testing

Activity	Duration	Deliverables	Tools
System Integration	2 weeks	Integrated system	Docker, Kubernetes
Unit Testing	2 weeks	Unit test coverage > 80%	Jest, JUnit
Integration Testing	2 weeks	API testing, Module integration	Postman, Cypress
End-to-End Testing	2 weeks	User flow testing	Cypress, Selenium
Performance Testing	2 weeks	Load testing, Stress testing	Apache JMeter, k6
Security Testing	2 weeks	Vulnerability assessment	OWASP ZAP, SonarQube
User Acceptance Testing	2 weeks	UAT sign-off	TestRail, Manual testing
Bug Fixing	2 weeks	Bug fixes, Polish	Jira

Team:

- QA Lead (1)
- QA Engineers (2)
- All Developers

Timeline: 8 weeks

Milestone: Testing complete

Testing Strategy:

1. Unit Testing

- Test individual components
- Jest for frontend
- JUnit/NestJS testing for backend
- Minimum 80% coverage

2. Integration Testing

- Test API endpoints
- Test module interactions
- Use test database
- Postman/Cypress

3. End-to-End Testing

- Test complete user flows
- Test critical paths
- Cypress for E2E

4. Performance Testing

- Load testing (10,000 concurrent users)
- Stress testing
- Apache JMeter/k6

5. Security Testing

- OWASP Top 10 compliance
- Penetration testing
- OWASP ZAP/SonarQube

14.7 Phase 6: Deployment & Launch

Activity	Duration	Deliverables	Tools
Infrastructure Setup	2 weeks	Cloud infrastructure	AWS/GCP/Azure, Terraform
CI/CD Pipeline	2 weeks	Automated deployment	GitHub Actions, Docker
Database Migration	1 week	Production database	PostgreSQL, Prisma
Server Configuration	1 week	Production servers	Nginx, Kubernetes
SSL Configuration	1 week	HTTPS setup	Let's Encrypt, Cloudflare
Domain Setup	1 week	Domain configuration	DNS providers
Staging Deployment	1 week	Staging environment	Docker, Kubernetes
Production Deployment	2 weeks	Production environment	Kubernetes, AWS/GCP

Data Migration	1 week	Data import (if any)	Custom scripts
Testing in Production	1 week	Smoke testing	Manual, Automated
Go-Live	1 day	Live website	Monitoring tools
Launch Monitoring	2 weeks	Monitoring and support	Datadog, Sentry

Team:

- DevOps Engineer (1-2)
- Backend Developer (1)
- Frontend Developer (1)

Timeline: 8 weeks**Milestone:** Platform launched

14.8 Phase 7: Maintenance & Enhancement

Activity	Duration	Deliverables	Tools
Bug Fixes	Ongoing	Stable platform	Jira, Sentry
Performance Monitoring	Ongoing	Performance reports	Datadog, New Relic
Security Updates	Ongoing	Security patches	Dependency scans
Content Updates	Ongoing	New content	CMS
Feature Enhancements	Ongoing	New features	Jira, Agile
User Support	Ongoing	User tickets	Support system
Analytics Review	Monthly	Analytics reports	Google Analytics

Team:

- Support Team (2-3)
- Development Team (Part-time)
- Content Team (1-2)

Timeline: Ongoing

14.9 Development Guidelines

14.9.1 Coding Standards

Frontend (Next.js/React/TypeScript):

- Use functional components with React Hooks
- Use TypeScript with strict type checking
- Follow ESLint rules
- Use Prettier for formatting

- Use named exports over default exports
- Use async/await over Promises
- Use React Query for server state

Backend (NestJS/TypeScript):

- Follow NestJS module structure
- Use DTOs for data validation
- Use class-validator for validation
- Use dependency injection
- Follow SOLID principles
- Use async/await
- Implement proper error handling

14.9.2 Git Branch Strategy

main

└── develop

| └── feature/module-name

| └── feature/feature-name

| └── hotfix/issue-number

| └── release/v1.0.0

Branch Naming:

- feature/feature-name
- bugfix/issue-number
- hotfix/issue-number
- release/v1.0.0

Commit Messages:

text

feat: Add user authentication

fix: Fix login validation bug

docs: Update API documentation

style: Format code with Prettier

refactor: Update user service structure

test: Add unit tests for auth module

chore: Update dependencies

14.9.3 Sprint Planning (Agile)

- **Sprint Duration:** 2 weeks
- **Sprint Planning:** Monday of Week 1
- **Sprint Review:** Friday of Week 2
- **Sprint Retrospective:** Friday of Week 2
- **Daily Standup:** 10 AM daily

14.9.4 CI/CD Pipeline

Git Push → GitHub Actions → Build → Test → Deploy to Staging → Approval → Deploy to Production

14.9.5 Recommended Libraries

Category	Libraries
UI Components	shadcn/ui, Radix UI, Tailwind UI
Forms	React Hook Form, Zod
Charts	Recharts, Chart.js
State Management	Zustand, React Query
Animations	GSAP, Framer Motion
3D	Three.js, Spline, React Three Fiber
HTTP	Axios, Fetch API
Validation	Zod, Class-Validator
Testing	Jest, Testing Library, Cypress
Date Handling	Day.js, date-fns
File Upload	React Dropzone, Multer
Real-time	Socket.IO
PDF Generation	jsPDF, React-PDF
QR Generation	qrcode.react
Video	Video.js, React Player

14.9.6 Team Responsibilities

Role	Responsibilities
Project Manager	Overall project coordination, Sprint planning, Stakeholder communication
Product Manager	Feature prioritization, Backlog management, Requirements
Solution Architect	Technical architecture, Technology selection, Best practices
Backend Lead	Backend architecture, Team management, Code review
Frontend Lead	Frontend architecture, Team management, Code review
AI Engineer	AI features, Model integration, API development

3D Developer	3D features, Three.js implementation, Optimization
QA Lead	Testing strategy, Quality assurance, Test automation
DevOps Engineer	Infrastructure, CI/CD, Deployment
UI/UX Designer	User interfaces, User experience, Prototyping
3D Designer	3D assets, Animations, Scene design

15. Appendices

15.1 Glossary

Term	Definition
Backend	Server-side components of the system
Frontend	Client-side components of the system
API	Application Programming Interface
Microservices	Architecture pattern with independent services
CI/CD	Continuous Integration / Continuous Deployment
Containerization	Packaging code and dependencies into containers
Orchestration	Managing containers and their interactions
Caching	Storing frequently accessed data for fast retrieval
Sharding	Splitting databases across multiple servers
Load Balancing	Distributing traffic across servers
Latency	Delay in system response
Throughput	Number of requests processed per second
Failover	Automatic switching to a backup system
Disaster Recovery	Recovery procedures after system failure

15.2 Technology Justification

Technology	Justification
Next.js	SSR for SEO, App Router for routing, React ecosystem, Built-in optimizations
NestJS	TypeScript-first, Modular architecture, Dependency injection, Scalable
PostgreSQL	Reliable, ACID compliance, JSONB support, Scalable
Three.js	Most mature WebGL library, Large community, Comprehensive features
GSAP	Professional animations, Browser compatibility, Performance
Docker	Containerization, Reproducibility, Ease of deployment

15.3 Design Decisions

Decision	Rationale
Monorepo vs Multi-repo	Monorepo for easier development and dependency management
SSR vs SPA	Next.js for SEO and performance benefits
REST vs GraphQL	REST for simplicity and caching

Custom Auth vs Auth0	Custom for full control and cost saving
SQL vs NoSQL	PostgreSQL for structured data and relationships
Self-hosted vs Cloud	Cloud for scalability and reliability

Conclusion

The Inquisitors Society 3D Intelligent Learning & Career Development Platform represents a complete architectural blueprint for transforming a student-run society into a full-scale digital ecosystem. Across this document, every layer of the system from the presentation layer and 3D rendering engine on the frontend, to the API gateway, business services, and AI provider factory on the backend has been defined with clear boundaries and responsibilities, so that development, testing, and deployment can proceed without ambiguity.

The layered approach adopted throughout (presentation → state/rendering → services → data access on the frontend; gateway → controllers → services → data access on the backend) ensures that each component can evolve independently. The security architecture enforces defense at every hop transport encryption, JWT authentication, RBAC authorization, and input validation before any data reaches storage. The AI provider factory pattern future-proofs the platform against vendor lock-in, allowing AgentRouter, OpenRouter, or Anthropic to be swapped or load-balanced without touching feature code.

Taken together, this SRS gives the development team, project supervisors, and future contributors a single source of truth: what the system does, how its pieces connect, and why each design decision was made. The next step is straightforward use this document as the baseline for sprint planning and begin implementation module by module, starting with authentication and the core LMS, since every other feature (internships, events, career tools, AI modules) depends on that foundation being solid.